ITA03-MOBILE COMPUTING
ASSIGNMENT-2

ASSIGNMENT 2 – CO2

CO2 Coverage: Mobile Telecommunication Systems; Multiple Access Techniques (SDMA, TDMA, FDMA, CDMA, OFDMA); GSM Architecture, Services, Radio Interfaces, Localization, Calling, Handover & Security; Advanced GSM Extensions (GPRS, HSCSD, DECT).

Nature of Assignment: Individual
Duration: 1 Week
Total Marks: 50
Submission Date: 12/11/2026

Q1 – Mobile Communication or Mobile Computing? 5 Marks Problem Context –

Field-Service Technology Audit

A utility company equips its field staff with: (a) a ruggedised tablet fetching work orders over cellular data, (b) a laptop plugged into Ethernet at different offices, (c) a fixed wireless router at a remote substation, (d) a push-to-talk handset, and (e) a tablet app that records meter readings offline and synchronises later.

Your Challenge

Investigate whether "wireless" and "mobile" mean the same thing, and whether mobile communication and mobile computing are the same thing.

Before analysis:

- Define at least four of your own classification criteria (e.g., user mobility, wireless link, tolerance of disconnection, location/context awareness, device resource limits).
- Predict how each of the five cases is classified (fixed-wired, nomadic, fixed wireless, mobile) and whether it is mainly a mobile-communication or a mobile computing problem, giving reasons.
- Design one additional ambiguous case of your own to test your criteria. **Test**

Your Reasoning

Build a small simulation of case (e) with link availability of 90%, 60% and 30%. Compare two strategies: send immediately (fail if disconnected) vs store-and-forward queue. Record updates delivered, delayed and lost, and mean delivery delay.

Inquiry

Which characteristics of mobile computing are exposed by your results that a wireless link alone does not solve? Which device categories need application-level changes?

Evidence Expected: Own classification criteria, reasoned predictions, an engineered ambiguous case, simulation counts for both strategies, and a justified conclusion.

Q2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic 6 Marks

Problem Context – Hostel-Campus Voice and Telemetry

A cell may use a 2 MHz band per direction. Option 1 is FDMA with 25 kHz channels. Option 2 is TDMA with 200 kHz carriers, each carrying 8 time slots. Both offer 80 nominal channels.

Your Challenge

Investigate whether equal nominal capacity means equal usable capacity.

- Estimate usable channels after guard bands, guard times, synchronisation and control overheads (state and justify your values).
- Predict which scheme gives better utilisation for (i) continuous voice, (ii) bursty data with about 10% duty cycle, and (iii) a 70/30 voice–data mix, and analyse why.

Design three contrasting traffic datasets, including one edge case of your own (e.g., very short calls or synchronised arrivals). Simulate dedicated-channel allocation under both schemes and record utilisation, blocking and wasted capacity.

Inquiry

Where do the two schemes diverge, and what causes it?

Design Decision

At a campus telemetry site, 90% of devices send a 200-byte report per minute. Is dedicated FDMA/TDMA channel allocation justified? Defend your answer.

Evidence Expected: Capacity calculations with overheads, reasoned predictions, three designed traffic conditions, utilisation/blocking evidence and a justified decision.

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem 7 Marks

Problem Context – Crowded Stadium Uplink

Several handsets transmit simultaneously in the same band to one base station, each spreading its data with its own code.

Your Challenge

Design a direct-sequence CDMA uplink simulation. Determine how data bits are spread, how the receiver despreads and decides, and what the processing gain is.

Before executing, predict the bit-error behaviour as the number of users grows for: (i) orthogonal (e.g., Walsh) codes with synchronous users, (ii) random non-orthogonal codes of the same length, and (iii) orthogonal codes where one user transmits at ten times the power. analyse your reasoning.

Implement and test with at least three code lengths (e.g., 8, 16, 64). Record bit errors over a stated number of bits, correlation values and processing gain.

Inquiry

- Why do orthogonal codes still fail when users are unsynchronised or powers are unequal?
- Why is power control essential in CDMA?
- Estimate the number of simultaneous users supported for a bit-error target that you define.

Evidence Expected: A working DS-CDMA simulation, predictions made before execution, bit-error evidence for three code lengths and three conditions, and a reasoned explanation of the results.

Q4 – OFDMA – Giving Each User the Subcarriers They Can Use Best 6 Marks

Problem Context – Lecture-Hall Downlink

Eight users share 64 subcarriers. Each user's channel quality varies across subcarriers (frequency-selective fading).

Your Challenge

- Before designing, determine how subcarrier spacing relates to symbol duration, why a cyclic prefix is needed, and verify orthogonality of at least three subcarriers numerically (FFT or inner product).
- Design three allocation approaches: fixed contiguous blocks, interleaved round-robin, and channel-aware best-subcarrier.
    - Predict total throughput and fairness (e.g., Jain's index) of each, with reasons.

Implement using generated fading gains with fixed seeds over at least 1,000 channel realisations. Record sum throughput, per-user throughput and fairness.

Inquiry

Why can maximising total throughput starve some users? How does the number of users change the benefit of channel-aware allocation?

Design Decision

Select an allocation for a lecture hall in which every student must receive at least a minimum rate that you define and justify.

Evidence Expected: Numerical orthogonality verification, predictions, three allocation schemes implemented, throughput and fairness results, and a justified allocation choice.

Q5 – SDMA – Reusing the Same Channel in Different Directions 6 Marks

Problem Context – Highway Cell With Directional Antennas

Vehicles along a highway use the same frequency and time slot. The base station can serve them with an omni antenna, a 120° sector antenna, or an N-element uniform linear array steered toward each user.

Your Challenge

- Derive the array factor of an N-element uniform linear array and predict how main-lobe width and side-lobe level change with N.
- Define an SIR threshold and predict the minimum angular separation at which two users can share the same resource.
- Compute patterns for N = 2, 4, 8, 16 and evaluate the SIR at user B when the beam is steered at user A, for separations of 5°–60°. Record beamwidth, side lobe level and the minimum separation meeting your threshold.

Inquiry

What limits SDMA in practice (angular spread, clustered users)? Why can two users in nearly the same direction not be separated?

Design Decision

At a toll plaza most vehicles arrive in the same direction. Would you rely on SDMA there? Defend your decision.
Evidence Expected: Array-pattern derivation, predictions, computed patterns and SIR results across angular separations, and a reasoned explanation of SDMA limits.

Q6 – Choosing Multiple-Access Techniques for Real Deployments 7 Marks Problem

Context – Four Deployments

Choose access methods for: (1) wide-area cellular voice, (2) a jam-resistant tactical link, (3) a high-rate stadium downlink, (4) a network of 5,000 low-rate sensors sending small reports.

Your Challenge

- Define at least five weighted criteria and justify the weights yourself. ● Rate each of SDMA, TDMA, FDMA, CDMA and OFDMA for each deployment using your own Q2–Q5 evidence wherever possible.
- Choose a primary technique and, where useful, a hybrid for each deployment. ● Perform a sensitivity check by changing weights by ±20%: does any choice change?

Inquiry

Why do real systems combine techniques (for example FDMA with TDMA)? Which of your choices is least robust, and why?

Evidence Expected: Independently justified criteria and weights, ratings traceable to your own Q2–Q5 evidence, sensitivity analysis and defended choices.

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate 13 Marks Review the evidence you generated

while working on Q1–Q6.

Select **ONE** solution that demonstrates a significant limitation as its problem size or operating requirement increases. Examples of limitations you may find in this variant:

- Dedicated FDMA/TDMA channels sit idle under bursty traffic.
- A CDMA link collapses under power imbalance or loss of synchronisation. ● Best-subcarrier OFDMA allocation starves weak users.
- SDMA fails when users are angularly clustered.

You are not permitted simply to replace the solution with a more advanced technology or standard and state that it is better. Your task is to understand the problem first and then develop an appropriate redesign.

A. Diagnose the Bottleneck 2 Marks

- Where most capacity, signalling or computational effort is being spent. ● Which counted quantity grows significantly as load or size increases. ● Whether resources are idle, wasted or repeated unnecessarily. ● Which characteristic of the original design causes the limitation.

Evidence Expected: A precise diagnosis supported by evidence from your original implementation and analysis.

B. Define the Design Objective 2 Marks

- State what your redesign must achieve before proposing it (e.g., reduce wasted capacity,

reduce signalling messages, avoid unnecessary resource holding, reuse information already available).

- Specify a measurable objective wherever possible.

Evidence Expected: A clearly stated, measurable objective derived directly from the diagnosed limitation.

C. Design the Revised Strategy 4 Marks

- What you have changed and what property or information the new design exploits.
- Which operations or resource usages are avoided or reduced. ● Whether additional state, preprocessing or complexity is introduced, and any new trade-offs.
- Provide pseudocode, a flowchart or a message-sequence design. ● Marks are based on the quality of reasoning, not merely on naming a known technique.

Evidence Expected: A coherent redesign directly addressing the bottleneck, with a design representation and technical justification.

D. Predict Before Implementing 1 Mark

- Predict how the redesigned solution should differ from the original and in which measure you expect improvement.

Evidence Expected: A clear, testable prediction made before experimental validation. **E.**

Implement and Validate the Redesign 2 Marks

- Implement a prototype or simulation and compare Original → Redesigned under the same or equivalent test conditions.
- Use counted measures (e.g., slots used, messages, blocking, bit errors, throughput, fairness); run-time is supporting evidence only.

Evidence Expected: Comparable experimental evidence for the original and redesigned solutions.

F. Defend the Final Design 2 Marks

- Was the prediction supported, and what was the magnitude of improvement? ● What new costs or trade-offs appeared, and when might the original still be preferable?
- A redesign that does not outperform the original is not automatically unsuccessful if you correctly analyse why the hypothesis failed.

Evidence Expected: A critical, evidence-based evaluation and a defensible final engineering conclusion.

ASSIGNMENT-2 – ANSWERS

Link Availability and Offline Operation

Planning assumptions used throughout the answers: The numerical values in the calculations are based on the assignment's stated data. Where the question does not specify a parameter, an explicit illustrative assumption is stated. Simulation result examples are labelled as illustrative and must not be presented as measured results unless the corresponding code is run.

ANSWER – Q1: Mobile Communication or Mobile Computing?

1. Classification criteria used

User mobility: whether the user/device changes physical location while working.

Connection type: wired, fixed wireless, or cellular/Wi-Fi wireless.

Disconnection tolerance: whether the task can continue without a network and synchronise later.

Context awareness: whether location, time, or changing network conditions affect the application.

Main purpose: communication transports voice/data; computing processes, stores, or acts on data.

2. Classification of the five cases

Case	Mobility / link classification	Main problem	Reason
(a) Rugged tablet using cellular work orders	Mobile; wireless	Both, mainly mobile computing	The tablet moves between field locations; the application must handle changing coverage and work-order data.
(b) Laptop plugged into Ethernet at different offices	Nomadic; fixed-wired while connected	Mainly mobile computing	The user moves between offices, but each session uses a wired link. Mobility is not the same as wireless access.
(c) Fixed wireless router at a remote substation	Fixed wireless	Mainly mobile communication/networking	The equipment is stationary; the radio link replaces a cable, but the

			device is not mobile.
(d) Push-to-talk handset	Mobile; wireless	Mainly mobile communication	The key service is real-time voice/group communication while the user moves.
(e) Tablet stores meter readings offline and synchronises later	Mobile; intermittently connected	Mainly mobile computing	Local storage, queueing, conflict handling, and delayed synchronisation are application responsibilities.

3. Additional ambiguous case

A technician carries a tablet around a factory, but it connects only to a wall-mounted Wi-Fi access point and saves inspection results locally when Wi-Fi is unavailable. It is mobile because the user moves, wireless when connected, and also a mobile-computing case because the application must tolerate outages.

4. Simulation assumptions and step-by-step calculation

Assume 100 meter updates are generated uniformly during a test. Link availability is the probability that an update finds the connection available. For the simple immediate-send strategy, a failed attempt is counted as lost. For store-and-forward, disconnected updates remain queued and are delivered at the next available opportunity. To make delay reproducible, assume a 1-minute check interval and that an update generated during an outage waits, on average, half of the unavailable portion of the interval. The table below is an illustrative model calculation, not a measurement from a physical network.

Immediate-send delivered = 100 × availability. Lost = 100 − delivered. Delayed = 0 because failed updates are discarded.

Store-and-forward eventually delivers all 100 updates if the queue has enough storage and connectivity returns; delayed updates = 100 × (1 − availability).

Illustrative mean delay for queued updates = (1 − availability) × 0.5 minute. This simple delay assumption can be replaced by measured delay in a real simulation.

Link availability	**Immediate delivered**	**Immediate lost**	**Queue delivered**	**Queue delayed**	**Illustrative mean queue delay / update**

90%	90	10	100	10	0.05 min (3 s)
60%	60	40	100	40	0.20 min (12 s)
30%	30	70	100	70	0.35 min (21 s)

5. Findings and conclusion

Mobile Computing Classification

A wireless link alone does not guarantee reliable application completion. Mobile computing also needs local storage, retry logic, timestamps, duplicate prevention, conflict resolution, battery-aware operation, and clear user feedback. Cases (a) and (e) need application-level changes; (b) may need session continuity and offline support when moving between offices. Case (c) primarily needs reliable fixed-wireless networking, while (d) needs handover, low latency, and voice-session continuity. Therefore, wireless and mobile are not synonyms, and mobile communication is a component of mobile computing rather than the same concept.

The classification criteria should be applied to the actual working arrangement rather than to the device name. A tablet can be used as a fixed terminal on a desk, carried between buildings, or operated in a vehicle. These situations create different mobility requirements even though the hardware is identical. User mobility describes whether the person changes location, while device mobility describes whether the device itself moves during the task. Connection mobility is a separate issue because a moving user may remain connected to one access point for a short period, or may need to transfer between several networks. Separating these ideas avoids treating every wireless device as automatically mobile.

Mobile Computing Classification

Mobile communication mainly concerns the transfer of information between endpoints over a communication link. Its important measures include signal strength, coverage, data rate, latency, packet loss, handover interruption and connection availability. Mobile computing includes the application and processing work performed by a portable or remotely used computing device. It also considers local storage, processing capability, battery use, user interface, synchronisation and consistency of data. A field-service tablet uses both areas: the radio network transports work orders, while the application displays them, records actions and preserves changes when the network is unavailable. A complete solution must therefore be evaluated at both levels.

The ruggedised tablet in case (a) is mobile because staff carry it while travelling between service locations. Its cellular connection supports mobile communication, and its work-order application makes it a mobile-computing system. The application should not assume that a connection is continuously available. It should retain the current work order, record the time and location of an update, and clearly show whether the update has reached the central server.

If the application reports success before the server confirms receipt, a technician may incorrectly believe that important maintenance information has been submitted.

Mobile Computing Classification

The laptop connected to Ethernet at different offices is best classified as nomadic rather than continuously mobile during the work session. The user changes location between offices, but the laptop is attached to a fixed wired connection at each location. Its network access is not wireless while connected through Ethernet. It may still use mobile-computing features such as portable processing and local files, but it does not require cellular handover while the cable is connected. This example demonstrates that mobility of the user, mobility of the device and wireless access are separate classification dimensions.

The fixed wireless router at a remote substation is a fixed-wireless communication installation. The radio link replaces a physical cable, but the router and service endpoint normally remain at a fixed location. The network must be designed for availability, antenna alignment, interference, weather exposure, power backup and backhaul reliability. It is not normally a mobile-computing problem because the endpoint is not expected to move with a user. However, engineers may remotely manage the router using mobile devices, which introduces a separate mobile-computing use case for the maintenance staff.

Mobile Computing Classification

The push-to-talk handset is a mobile communication device because the user carries it and depends on a radio connection to communicate while moving. Its most important service requirements are rapid call setup, group communication, intelligible audio, coverage and continuity during movement. A short interruption can be more serious than a small reduction in peak data rate because the user may be coordinating a safety-critical field activity. The handset may contain computing functions such as contact lists and location display, but its primary purpose in this scenario is communication. This distinction helps select appropriate performance measures for the system.

The offline meter-reading application is a clear mobile-computing example because the application must continue to perform useful work when the network is disconnected. A robust design stores readings in a local database, assigns each record a unique identifier, records a timestamp and synchronises pending records when service returns. The application should preserve the original reading if a retry occurs, so that the same record is not inserted twice. It should also report the status of each item as pending, synchronised or requiring attention. These behaviours are application responsibilities and cannot be provided by the radio link alone.

Link Availability and Offline Operation

The link-availability experiment should distinguish successful immediate delivery from eventual delivery. At 90% availability, immediate transmission delivers approximately 90 of 100 updates and loses 10 if no retry is attempted. At 60% availability, the corresponding values are 60 delivered and 40 lost; at 30%, only 30 are delivered and 70 are lost. A store-and-forward design can retain the updates generated during outages and deliver them after reconnection, provided that the queue does not overflow and the server accepts delayed

records. The experiment should therefore report both delivery ratio and delivery delay rather than using one metric alone.

A realistic simulation should also consider the pattern of disconnections. Two systems can have the same 60% availability but different user experiences. One may have short, frequent interruptions, while another may remain disconnected for a long continuous period. The second case creates a larger queue and may delay urgent records for longer. A useful test should vary outage duration, reconnection interval, packet size and retry policy while keeping the generated records identical between strategies. The output should include generated records, immediately delivered records, queued records, eventual deliveries, duplicates, losses and maximum queue length. These measurements explain the behaviour more clearly than availability percentage alone.

Mobile Computing Classification

The final classification should be used to guide engineering decisions. Communication engineers should verify coverage, radio quality, access delay and handover behaviour. Application developers should verify offline operation, local data protection, synchronisation, conflict handling and clear status messages. Device designers should consider battery life, ruggedness, storage capacity and processing performance. For the utility company, the most reliable solution is a coordinated design in which the network provides connectivity when possible and the application remains useful when connectivity is absent. This is the central difference between merely providing a wireless connection and delivering a dependable mobile-computing service.

Data consistency is a major issue when a field worker edits information offline. The central server and the tablet may both contain a version of the same work order. If another employee changes the work order before the tablet reconnects, the system must decide how to combine the updates. Some fields can be merged automatically, while others require a supervisor to review the conflict. For example, a timestamped meter reading should normally be retained as a separate observation rather than overwritten by a later value. The application should preserve an audit trail so that every change can be traced to its source.

Link Availability and Offline Operation

Security must be considered even when the device is disconnected. A lost tablet may contain customer details, equipment locations and maintenance records. The application should protect stored data using device authentication and encryption, and it should avoid exposing sensitive information on the lock screen. When a connection becomes available, synchronisation should use an authenticated and encrypted session. The system should also be able to revoke access for a device that has been reported lost. Offline operation should not mean that local information is left unprotected until the next connection.

The user interface should make network state visible without interrupting the technician's work. A clear status such as connected, synchronising, pending or failed helps the user understand whether an update has reached the server. A background retry process should use controlled intervals rather than continuously attempting to reconnect, because repeated attempts consume battery power and network resources. The application should also allow the user to continue recording essential information while a synchronisation operation is running. This makes the service more dependable in locations with variable coverage.

Link Availability and Offline Operation

The simulation can be extended by modelling a finite local queue. If a device generates one record per minute and remains disconnected for 30 minutes, it must store at least 30 records plus a safety margin. If each record occupies 2 kB after metadata, the raw queue requirement is approximately 60 kB, excluding database overhead. The calculation should be repeated for the longest expected outage and the maximum number of records generated during that period. A queue that is too small can turn a temporary network outage into permanent data loss.

A useful performance measure is eventual delivery probability, defined as the number of generated records eventually accepted by the server divided by the total number generated. This differs from immediate delivery probability, which counts only records delivered during the first attempt. A store-and-forward system may have a low immediate delivery rate but a high eventual delivery rate. Both values should be reported with average and 95th-percentile delay. This makes the trade-off between availability and timeliness visible and prevents the application from appearing reliable simply because it stores every record without delivering it promptly.

Mobile Computing Classification

The additional ambiguous case can be evaluated using the same criteria as the five original devices. The factory technician carries a tablet, so the user and device are mobile. The tablet uses Wi-Fi at a fixed access point when available, so the connection is wireless but not necessarily cellular. It continues collecting inspection data during an outage, so it is a mobile-computing application with disconnection tolerance. This example confirms that a device can be mobile even when its network access is local Wi-Fi, and that mobile computing is not restricted to cellular networks.

For field deployment, coverage measurements should be collected along the actual routes used by technicians rather than only at office locations. A route survey can identify dead zones, weak indoor coverage and locations where handover occurs. The application team can use these results to decide how long records may remain queued and whether certain tasks require an online connection. Network measurements should be correlated with application logs so that a failed transaction can be traced to poor radio coverage, server unavailability or an application error. This joint diagnosis avoids blaming the wrong subsystem.

Mobile Computing Classification

The classification framework can be applied to future devices without changing its basic structure. A wearable scanner, vehicle-mounted terminal or remote camera can be evaluated by asking whether the user or device moves, what type of link is used, how the application behaves during disconnection and whether context affects the task. The result should identify the main engineering responsibility: radio communication, mobile application design or a combination of both. This approach is more useful than memorising fixed labels because new devices often combine wired, wireless, fixed and mobile functions.

The field-service system should define what happens when two records conflict during synchronisation. A simple last-write-wins rule may be suitable for a technician's status note, but it may be unsafe for a meter reading or a completed safety inspection. The application can preserve both versions and request a review when the values cannot be merged confidently.

Every record should contain a unique identifier, device identifier, timestamp and version number. These fields help the server distinguish a new observation from a retransmission and maintain a reliable history of field activity.

Link Availability and Offline Operation

The network should not be considered available only because the device is associated with an access point. A connection may exist while the server is unreachable, DNS resolution fails or an application request times out. The application should verify successful server acknowledgement before marking a record as synchronised. It should also handle partial uploads and interrupted sessions without corrupting the local record. This layered view of availability is important because radio connectivity, IP connectivity and application-service availability are different conditions and should be logged separately.

A field worker may move through areas with different network technologies, such as cellular coverage outdoors and Wi-Fi inside a maintenance building. The application should maintain a consistent session when the underlying connection changes, or it should safely resume after a temporary interruption. Network switching may change latency, address information and throughput. The application should avoid assuming that the same connection properties remain constant throughout a task. This is another reason why mobile computing requires more than simply establishing a wireless link.

Mobile Computing Classification

The classification results can be represented as a decision process. First determine whether the user or endpoint changes physical location. Next identify whether the access link is wired, fixed wireless or mobile wireless. Then examine whether the application must continue during disconnection and whether it uses location or context. Finally identify whether the main requirement is communication transport, local computing or both. Applying the questions in the same order makes the classification repeatable and reduces confusion when a device has more than one operating mode.

The evaluation should also consider the consequences of failure. A delayed work order may inconvenience a technician, while a lost safety inspection or incorrect meter reading may have a serious operational impact. Critical records should receive stronger persistence, validation and retry handling. The application can use different priorities for routine updates and urgent alarms. The network can then reserve or prioritise resources for important traffic. This illustrates how the communication and computing layers cooperate to deliver a service that meets the actual business requirement.

Mobile Computing Classification

The mobile-computing design should support a clear recovery process when a device is replaced. If a tablet is damaged, pending records should be recoverable through an authenticated backup or a controlled transfer to another device. The server should be able to identify which records have already been accepted so that the replacement device does not upload duplicates. A recovery test should include device loss, account reauthentication and restoration of pending work. This is an application continuity requirement that is separate from radio coverage.

Location awareness should be used only where it supports the task. A meter-reading application may attach a location or asset identifier to confirm that the technician visited the correct site. Location data can also help the operator plan routes and investigate incomplete work. However, inaccurate positioning indoors can create false confidence. The application should allow a user to confirm the asset manually when location accuracy is insufficient, and it should record the source and accuracy of location information when that information affects a business decision.

Link Availability and Offline Operation

The final service assessment should combine network and application indicators. Network indicators include availability, signal quality, latency and handover interruption. Application indicators include successful record creation, eventual synchronisation, duplicate rate, queue overflow and user-visible failure. A system may have excellent radio coverage but still fail because the application loses data during synchronisation. Conversely, a store-and-forward application may complete the task successfully despite temporary coverage loss. Both layers must be measured to judge the real service quality.

The application should support a clear user workflow for work completed while offline. The technician should be able to review pending entries, correct an error before synchronisation and see which records require attention. If the server rejects a record because of an invalid asset identifier, the application should explain the reason and preserve the original entry for correction. Silent failure is particularly harmful in field-service systems because the user may leave the site believing that the task has been completed successfully.

Mobile Computing Classification

The field-service example demonstrates that mobile computing is a combination of device, application, network and operating procedure. A good design considers how the technician works, where the device is used, what happens when the network fails and how the organisation verifies completed work. The classification criteria help identify these requirements before selecting a communication technology. This avoids treating wireless connectivity as the complete solution when application reliability and data integrity are equally important.

ANSWER – Q2: FDMA vs TDMA Under Different Traffic

1. Nominal capacity calculation

Total bandwidth per direction = 2 MHz = 2,000 kHz.

FDMA nominal channels = 2,000 kHz ÷ 25 kHz/channel = 80 channels.

TDMA nominal carriers = 2,000 kHz ÷ 200 kHz/carrier = 10 carriers.

TDMA nominal time slots = 10 carriers × 8 slots/carrier = 80 slots.

2. Assumed overhead and usable capacity

The question does not provide exact guard-band, guard-time, synchronisation, or control values. For a transparent comparison, assume FDMA loses 8% of nominal capacity to guard bands/control, while TDMA loses 10% to guard time, synchronisation, and control. These are assignment assumptions, not standard-defined values.

FDMA usable channels = $80 \times (1 - 0.08) = 73.6 \approx 74$ channels.

TDMA usable slots = 80 × (1 − 0.10) = 72 slots.

Although both begin at 80 nominal channels, usable capacities differ because their overheads differ.

Scheme	**Nominal capacity**	**Assumed overhead**	**Estimated usable capacity**
FDMA	80 channels	8%	80 × 0.92 = 73.6 ≈ 74
TDMA	80 slots	10%	80 × 0.90 = 72

3. Traffic predictions

Traffic condition	**Expected better utilisation**	**Reason**
Continuous voice	FDMA or TDMA, depending on implementation	Voice needs a steady rate; dedicated FDMA channels avoid slot scheduling, while TDMA can share carrier hardware efficiently.
Bursty data, 10% duty cycle	TDMA if slots can be reassigned; otherwise neither is efficient	A dedicated channel sits idle about 90% of the time when the source is inactive.
70% voice / 30% data	TDMA with scheduling, or a hybrid	Voice can receive regular slots while data uses available slots; strict fixed allocation wastes capacity when traffic varies.

4. Three designed traffic datasets and illustrative allocation results

For a small reproducible comparison, assume 72 usable resource units for TDMA and 74 for FDMA. Each active call/report occupies one resource unit. Utilisation = occupied resource units ÷ usable units × 100%. Blocking = requests rejected because all units are occupied. The figures below are illustrative scenario counts chosen to demonstrate the method; replace them with counts from an actual simulation before claiming measured results.

Dataset	**Offered load pattern**	**FDMA utilisation / blocked**	**TDMA utilisation / blocked**	**Interpretation**

A: steady voice	60 continuously active calls	60/74 = 81.1%; 0 blocked	60/72 = 83.3%; 0 blocked	Both work well for steady demand.
B: bursty telemetry	8 active reports out of 80 registered devices at a time	8/74 = 10.8%; 0 blocked	8/72 = 11.1%; 0 blocked	Most dedicated capacity remains idle.
C: short synchronised burst	80 requests arrive together	74/74 = 100%; 6 blocked	72/72 = 100%; 8 blocked	A sudden peak exceeds available resources; capacity and admission policy matter.

5. Campus telemetry decision

Report size = 200 bytes × 8 = 1,600 bits per report.

At one report per minute, average payload rate per device = 1,600 ÷ 60 = 26.67 bit/s, before headers and retransmissions.

For 5,000 devices, aggregate payload rate = 5,000 × 26.67 = 133,350 bit/s ≈ 133.35 kbit/s.

Only 10% of devices transmit in a given instant if activity is asynchronous; this is still a low-rate, bursty traffic pattern compared with continuously reserved voice channels.

Data Synchronisation and Consistency

Conclusion: Dedicated FDMA/TDMA allocation is generally not justified for thousands of low-rate telemetry devices because resources would remain idle most of the minute. A scheduled or contention-based shared channel, with batching and retry control, is more suitable. Dedicated resources remain useful for continuous voice or latency-sensitive traffic.

FDMA and TDMA divide a shared radio resource in different dimensions. FDMA assigns a separate frequency channel to each active connection, so users transmit at the same time but occupy different frequency bands. TDMA assigns users different time slots on a carrier, allowing several users to share the same carrier at different instants. Both methods require coordination so that one user's transmission does not overlap another user's allocated resource. The nominal channel count is only a first estimate; guard bands, guard times, control signalling, synchronisation and practical operating margins reduce the number of resources that can carry useful traffic.

Data Synchronisation and Consistency

The FDMA calculation begins with a total bandwidth of 2 MHz, or 2,000 kHz, per direction. If each channel occupies 25 kHz, the nominal number of channels is 2,000 divided by 25, which gives 80. This value assumes that the entire stated bandwidth can be divided into equal

channel widths. A practical frequency plan may reserve edge spectrum, control channels or separation between adjacent allocations. The usable number must therefore be calculated after the actual system constraints are stated. The assignment's 8% overhead is an illustrative assumption for comparison and should not be described as a universal FDMA standard value. For TDMA, a 200 kHz carrier is divided into eight time slots. Ten such carriers fit into 2,000 kHz, giving 80 nominal slots per frame across the ten carriers. A slot is not an independent frequency channel; it is a time resource on a carrier. The system needs timing advance or equivalent timing control so that bursts from different mobiles arrive in their expected positions. Guard time protects against timing uncertainty and propagation differences. Synchronisation and control information also consume part of the frame, which explains why the number of usable traffic slots is lower than the nominal total.

Simulation Design and Validation

The overhead assumptions produce approximately 74 usable FDMA channels and 72 usable TDMA slots. These rounded values are suitable for the illustrative allocation experiment, but a real design should calculate overhead from the exact frame structure and channel specification. Rounding must be performed carefully because a fraction of a channel cannot normally be assigned as a complete traffic resource. If the calculation gives 73.6 usable channels, the conservative integer capacity is 73 or 74 only when the system's allocation rules justify the rounding. The selected rounding convention should be stated and used consistently in every scenario.

A fair comparison requires the same offered traffic sequence for FDMA and TDMA. Each request should have an arrival time, service duration, traffic type and priority. The simulator should process requests in chronological order, allocate a free resource when one exists, and record blocking when all eligible resources are occupied. For TDMA, the simulator should also account for the fact that a user may be assigned a slot periodically rather than continuously. If the model treats every slot as an immediately available independent channel, it should state that this is a simplified capacity model and not a full frame-level timing simulation.

Capacity and Resource Utilisation

Under low and steady traffic, both methods may show low blocking because most resource units remain available. As the offered load approaches the usable capacity, blocking rises rapidly. This behaviour is expected in a finite-resource system: the final few free channels become increasingly likely to be occupied when new requests arrive. A useful report should plot blocking probability against offered traffic rather than reporting only one operating point. It should also include utilisation, carried traffic, rejected requests and average waiting time where waiting is permitted. These measures show whether a method is efficient and whether it meets the service requirement.

The difference between continuous voice and intermittent telemetry is important. A voice call typically requires a resource throughout the call, even when the speaker is briefly silent. A meter-reading device may transmit only a short report once per minute and remain idle for most of the interval. Permanently assigning one channel to every sensor would reserve resources that are unused for most of the time. A shared scheduled system can allow many sensors to use a common pool, provided that access requests, collisions, scheduling delay and

retransmissions are controlled. Resource assignment should match the traffic pattern rather than treating all devices as continuous sources.

Simulation Design and Validation

For 5,000 telemetry devices, a 200-byte report contains 1,600 payload bits. One report per minute gives an average payload rate of about 26.67 bit/s per device and approximately 133.35 kbit/s across all devices. This average does not mean that the system needs a continuous 133.35 kbit/s allocation for every sensor. Reports may be distributed over time, aggregated or scheduled in short bursts. The actual radio demand will be higher than the payload rate because headers, acknowledgements, retransmissions and access procedures add overhead. A simulation should therefore distinguish application payload from total transmitted bits.

A bursty-arrival experiment should compare at least three traffic patterns: evenly spaced arrivals, independent random arrivals and a synchronised burst in which many devices report together. Equal average traffic can produce very different blocking and delay under these patterns. The synchronised case is particularly important for utility systems after a power restoration or network outage, when many meters may attempt to reconnect at the same time. Randomising the reporting time or applying a backoff procedure can spread the load. The model should record the peak number of simultaneous requests as well as the average offered traffic.

Data Synchronisation and Consistency

The appropriate decision is to use dedicated or guaranteed resources for continuous voice and latency-sensitive services, while using scheduled or contention-based shared access for low-rate telemetry. This is not a claim that TDMA is always superior to FDMA or that one access method is universally best. The choice depends on traffic duration, synchronisation requirements, spectral conditions, equipment complexity and service guarantees. A hybrid network can reserve predictable resources for voice while allowing intermittent devices to share the remaining capacity. The design should be accepted only after measured blocking, delay and delivery ratio meet the application targets.

The Erlang concept can be used to describe offered traffic when the service consists of calls that occupy a resource for a measurable duration. One Erlang represents one resource continuously occupied for the observation period. Offered traffic depends on the arrival rate multiplied by the average holding time. If call arrivals increase or average call duration becomes longer, the offered load increases even if the number of subscribers does not change. This explains why subscriber count alone is not enough to predict blocking. The traffic model must include busy-hour activity and holding-time assumptions.

Queue Management and Recovery

For a loss system with no waiting queue, a request is blocked when every usable resource is occupied. The Erlang-B formula estimates the blocking probability for a specified offered load and number of resources under its assumptions. The formula is useful for voice traffic because a caller normally expects an immediate connection rather than waiting in a queue. It should not be applied blindly to short telemetry packets, which may be buffered and retried. A

telemetry model should represent packet arrivals, queueing and deadlines, while a voice model can use a loss-system approximation when its assumptions are appropriate.
The FDMA and TDMA comparison should include the effect of holding time. If a voice call lasts several minutes, its assigned resource remains occupied for that duration. A short telemetry message may occupy a slot for only a small fraction of a second. Treating both as identical one-unit calls exaggerates the resource requirement for telemetry. A more realistic simulator should convert packet size and transmission rate into airtime, add protocol overhead and then calculate the resource occupancy. The resulting comparison will better reflect the difference between continuous voice sessions and intermittent data reports.

Capacity and Resource Utilisation

Guard bands and guard times perform similar protective roles in different dimensions. A guard band provides frequency separation between adjacent channels to reduce leakage and interference from imperfect filters or oscillators. A guard time provides temporal separation between bursts so that timing uncertainty and propagation delay do not cause one user's transmission to overlap another's. Increasing either guard interval improves robustness but reduces the fraction of the resource available for payload. The selected values should be justified using the actual radio system and not chosen only to produce a preferred capacity result.
Frequency reuse is another important factor in an FDMA network. The same frequency channel can be assigned to multiple cells if the separation is sufficient to keep co-channel interference within acceptable limits. A smaller reuse distance increases the number of times a channel can be reused but also raises interference risk. TDMA systems can use the same frequency-reuse principles because time slots are arranged on carriers that still experience radio propagation and interference. The access method and the frequency plan are related but distinct parts of network design.

Data Synchronisation and Consistency

The synchronised-burst dataset should be treated as a stress test rather than an average operating condition. If all sensors transmit immediately after a common event, the access channel may experience a sudden surge of requests. A random backoff can spread attempts over time, but it may also increase delivery delay. A scheduled system can assign different reporting windows to groups of devices before the event occurs. The report should compare peak collision probability, mean delay, 95th-percentile delay and the number of reports delivered before the deadline. These measures reveal whether the system recovers after the burst.
A hybrid resource plan can reserve a predictable fraction of capacity for voice and use the remaining capacity for shared data access. The reserved portion protects voice quality during high traffic, while the shared portion can be dynamically allocated to telemetry. The split should not be fixed without evidence; it should be based on the measured busy-hour voice load and the expected data burst. A controller may adjust the data pool within safe limits when voice demand changes. The design must also prevent data traffic from consuming resources needed for emergency communication.

Simulation Design and Validation

Energy consumption differs between access methods. A TDMA device may need to wake only for its assigned slot, which can support sleep scheduling and battery savings. FDMA receivers may need to monitor a dedicated channel continuously depending on the system design. Shared access can reduce reserved resources but may require devices to listen for scheduling grants or retry after collisions. The energy model should include transmit time, receive time, idle listening, synchronisation and retransmission. For battery-powered sensors, the best radio allocation is not necessarily the one with the highest raw spectral efficiency.

A reproducible comparison should publish the arrival-generation method and all random seeds. The same traffic sequence should be replayed under FDMA and TDMA, and any difference in resource occupancy should be caused by the allocation method rather than by a different random workload. The code should log each allocation decision and the reason for blocking. It should also distinguish a resource that is unavailable because it is occupied from one that is unavailable because of interference or control overhead. This level of detail makes the results easier to audit and reproduce.

Data Synchronisation and Consistency

The conclusion should avoid claiming that one method always provides more capacity. The calculated 74 FDMA channels and 72 TDMA slots depend on the stated bandwidth, resource widths and assumed overhead. Different standards and frame structures can produce different results. The meaningful finding is that both methods partition a fixed resource and that the traffic pattern determines how efficiently the partition is used. A deployment should select a resource structure that matches the required service, then validate blocking, delay, fairness and energy use using representative traffic.

The resource model should distinguish between the number of simultaneous users and the number of subscribers registered in the network. A large subscriber population may generate only a modest number of simultaneous voice calls during off-peak periods. During the busy hour, the active fraction and call duration determine the offered traffic. For telemetry, the active fraction is often much smaller but can rise sharply after a common event. A useful capacity report should therefore state the subscriber count, activity probability, arrival rate, average service time and peak synchronisation assumption.

Simulation Design and Validation

The blocking calculation should be checked against the physical resource count. If a system has 74 usable channels, it cannot carry more than 74 simultaneous full-rate calls under the one-channel-per-call assumption. The actual blocking probability depends on offered traffic and the statistical arrival and holding-time model. A simulation should compare its measured blocking with the Erlang-B estimate for voice-like traffic. Differences may arise from finite simulation duration, non-Poisson arrivals, non-exponential holding times or an implementation error. The assumptions should be listed before the two results are compared.

The TDMA frame structure introduces scheduling delay that is not captured by a simple count of 72 slots. A user may have to wait until its assigned slot appears in the next frame. If the frame repeats regularly, the maximum scheduling wait depends on the frame duration and slot position. A packet that spans several slots may require multiple allocations or fragmentation.

The simulator should include these effects when the goal is to estimate packet delay rather than only nominal capacity. This distinction is especially important for real-time traffic.

Data Synchronisation and Consistency

Frequency stability affects both FDMA and TDMA. In FDMA, oscillator error can shift energy toward adjacent channels and increase interference. In TDMA, timing and frequency errors can cause bursts to overlap or drift from their expected positions. Devices therefore require suitable frequency references, synchronisation procedures and guard intervals. These requirements add hardware and signalling costs. A complete comparison should consider not only spectrum occupancy but also the complexity required to maintain clean separation between users.

The telemetry system can reduce burst pressure by assigning different reporting offsets to groups of sensors. If devices report at random times within a reporting window, the number of simultaneous requests is lower than if all devices transmit at the same instant. The offset can be generated when a device is commissioned or updated by the network. The system should still retain a backoff mechanism because power restoration or network recovery can cause many devices to lose their schedule and reconnect together. Staggering improves average behaviour but does not remove the need for overload protection.

Capacity and Resource Utilisation

The resource plan should also include a reserve for maintenance and unexpected load. Operating every channel at full occupancy leaves no margin for traffic bursts or equipment failure. A small reserve can improve service continuity, although it reduces the number of resources available for normal traffic. The reserve should be selected using measured busy-hour demand and the required blocking target. The report should state whether reserved resources are included in the usable channel count or excluded before the capacity calculation to avoid counting them twice.

When comparing the access methods, the same service-quality target should be applied to both. For voice, this may include a blocking probability and a maximum setup delay. For telemetry, it may include a delivery ratio and reporting deadline. A method should not be declared superior because it carries more requests if those requests are delivered too late or with unacceptable reliability. The evaluation should compare the number of successfully completed services, not merely the number of nominal channels or slots.

User Interface and Network Status

The operator should review the traffic profile periodically. A campus network may have different patterns during teaching hours, examinations, holidays and large events. A fixed allocation based on one average day can leave capacity unused at some times and cause congestion at others. Traffic measurements can be used to adjust scheduling windows, reserve levels and the division between voice and data resources. This operational feedback is important because a resource plan is not a one-time calculation; it must adapt as user behaviour changes.

The traffic model should separate offered traffic from carried traffic. Offered traffic represents all requests presented to the system, while carried traffic represents requests that successfully obtain resources. The difference is caused by blocked or rejected requests. Reporting only

carried traffic can make an overloaded network appear stable because the blocked demand is hidden. A useful results table should show offered load, carried load, blocking probability, utilisation and the number of requests rejected for each test condition.

Simulation Design and Validation

The final access-method recommendation should be reviewed against both average and peak demand. FDMA and TDMA can provide predictable resource separation, but fixed allocation may not adapt efficiently when traffic is uneven. Shared scheduling can improve utilisation for intermittent reports, but it requires control signalling and queue management. A hybrid arrangement can preserve predictable resources for voice while allowing low-rate devices to share unused capacity. The selected design should be validated with the same traffic traces and the same service targets.

ANSWER – Q3: CDMA, Spreading and the Near–Far Problem

1. DS-CDMA operation

Map each data bit to bipolar form: bit 1 → +1 and bit 0 → −1.

Assign each user a spreading code of length L, containing +1 and −1 chips.

Spread one data bit by multiplying it by all L chips of that user's code. One bit therefore becomes L chips.

Add the chip sequences from simultaneous users to form the received signal.

At the receiver, multiply the received chips by the desired user's code and average/sum across L chips. A positive result decides bit 1; a negative result decides bit 0.

2. Processing gain calculation

Processing gain in linear form is Gp = chip rate ÷ bit rate = code length L. In decibels, Gp(dB) = 10 log10(L).

Code length L	**Linear processing gain**	**Step-by-step dB calculation**	**Approximate gain**
8	8	10 log10(8)	9.03 dB
16	16	10 log10(16)	12.04 dB
64	64	10 log10(64)	18.06 dB

3. Correlation and predictions before testing

For two code vectors a and b, normalised correlation is ρ = (a·b)/L. Orthogonal codes have a·b = 0, so ρ = 0 when aligned and synchronised. Random codes usually have non-zero correlation, causing multi-user interference.

Condition	**Prediction as user count grows**	**Reason**

Synchronous orthogonal codes, equal power	Lowest bit-error rate while code orthogonality is preserved	Cross-correlation ideally equals zero.
Random non-orthogonal codes	Bit errors increase sooner	Non-zero cross-correlation adds interference to the desired user's despread signal.
Orthogonal codes, one user at 10× power	Near–far interference may cause errors for weaker users	Timing errors, imperfect channel estimation, and non-ideal orthogonality let strong-user energy leak into other users.

4. Small reproducible simulation plan

Use a fixed random seed (for example 42), 10,000 information bits per user, and code lengths L = 8, 16, 64. Run each of the three conditions separately. Record total transmitted bits, error bits, BER = errors ÷ tested bits, code cross-correlations, and processing gain. For a fair comparison, keep the same bit sequence and noise setting between the original and modified cases.

5. Worked correlation example

Let a = [+1, +1, −1, −1] and b = [+1, −1, +1, −1]. Their dot product is (1×1) + (1×−1) + (−1×1) + (−1×−1) = 1 − 1 − 1 + 1 = 0. Therefore ρ = 0/4 = 0: these two codes are orthogonal when aligned.

6. Capacity estimate and conclusion

Simulation Design and Validation

Define the target as BER ≤ 1%. The maximum simultaneous users cannot be calculated from code length alone; it depends on received SNR, synchronisation, code cross-correlation, channel conditions, and power control. A defensible experiment should increase users one at a time and report the largest count for which measured BER remains ≤ 1%. Power control is essential because the base station should receive users at approximately comparable power; otherwise, a nearby strong user can overwhelm a distant weak user.

Direct-sequence CDMA allows several users to transmit over the same frequency band at the same time by assigning different spreading codes. Each information bit is represented by a sequence of chips, and the receiver correlates the received signal with the code assigned to the intended user. The desired user's contribution adds coherently during despreading, while signals using sufficiently uncorrelated codes contribute less to the decision statistic. This is the basic reason that code separation can support simultaneous users. The method does not eliminate interference; its performance depends on code properties, timing, received powers and the radio channel.

Data Synchronisation and Consistency

The bipolar mapping converts binary data into values of positive one and negative one. A spreading code is also represented by positive and negative chips. Multiplying the data symbol by every chip produces a longer chip sequence. If the spreading length is L, one data bit is represented by L chips, so the chip rate is L times the information bit rate for a fixed symbol duration. The receiver performs the reverse correlation operation. The processing gain describes the spreading ratio, but it should not be interpreted as an automatic increase in transmitted energy or as a guarantee of a particular number of users.

For L equal to 8, 16 and 64, the ideal processing gains are 9.03 dB, 12.04 dB and 18.06 dB respectively. These values follow from 10 log10(L). A longer code offers greater processing gain in the simplified model, but it also increases chip rate or requires a longer symbol interval if the bit rate is held fixed. A real system must consider bandwidth, synchronisation and delay constraints. The gain is useful for distinguishing a spread signal from some interference, but the achievable capacity still depends on the total interference level and the required bit-error rate.

Data Synchronisation and Consistency

Orthogonality is a property of the cross-correlation between code sequences under a specified timing relationship. Two codes with a zero dot product have zero ideal cross-correlation when they are aligned and evaluated over the same code interval. If one signal arrives with a timing offset, the receiver may no longer observe the ideal zero value. Multipath propagation can create delayed copies of a signal, and those copies can also reduce orthogonality. Therefore, the statement that two codes are orthogonal must include the assumptions of alignment, synchronisation and the channel conditions under which the correlation is measured.

The near–far problem occurs when a strong user signal reaches the base station at much greater power than a weak user's signal. The strong signal can create interference that remains significant after the receiver attempts to despread the weaker user's transmission. This can happen even when the spreading codes have good correlation properties. Power control addresses the problem by adjusting mobile transmit powers so that received powers are kept within a controlled range. Open-loop power control can provide an initial estimate, while closed-loop control can make corrections based on measurements from the network.

Simulation Design and Validation

A useful simulation should first test a clean synchronous case with equal received powers. It should then introduce random non-orthogonal codes, timing offsets, additive noise and unequal received powers. Each modification should be applied separately before combinations are tested, because this makes the cause of a BER change easier to identify. The same information bits and random seed should be used for each comparison. Report the number of transmitted bits, error bits, BER, average cross-correlation and received power ratio. Without these quantities, a statement that one code set is better is difficult to verify.

The worked code example demonstrates how correlation is calculated. The dot product of the two four-chip vectors is zero, so their normalised correlation is also zero when they are aligned. A larger code family should be checked pair by pair rather than assuming that all different-looking sequences are orthogonal. For a set of K codes, calculate the cross-correlation for every distinct pair and record the maximum absolute value. This maximum is

often more informative than the average because a single highly correlated pair may cause unacceptable interference when both users are active.

Simulation Design and Validation

The maximum number of simultaneous users should be found experimentally for a clearly stated quality target. If the target is BER no greater than one percent, increase the active-user count gradually and record the BER for each count over enough transmitted bits. The capacity estimate is the largest tested user count that meets the target under the selected channel and power assumptions. Repeat the experiment for multiple noise levels and power-control errors. A single result from one random sequence is not sufficient because BER is a statistical quantity and can vary between trials.

CDMA also needs practical synchronisation and receiver processing. The base station must estimate timing, channel response and received power, then maintain these estimates as users move. Multipath combining can improve reception when delayed signal components are useful, but inaccurate channel estimates can cause residual interference. The receiver's computational complexity increases with the number of users and the sophistication of the detection method. These implementation costs should be considered alongside theoretical processing gain. A method that performs well in an ideal correlation model may be unsuitable if the required receiver cannot be implemented within power or cost limits.

Interference and Signal Quality

The crowded-stadium uplink is therefore a useful environment for examining CDMA because many users may transmit simultaneously and received powers can differ substantially. The system should combine code assignment with power control, admission control and interference monitoring. If the measured BER rises above the target, the controller can reduce the number of active users, adjust power, select a different code or schedule some transmissions. The final conclusion should state the conditions under which CDMA is effective and identify the limits caused by near–far effects, code correlation, timing error and finite receiver capability.

The received CDMA signal can be represented as the sum of the desired user's spread waveform, the spread waveforms of other users and noise. Despreading multiplies the received sequence by the desired code and combines the chips over the bit interval. The desired signal adds in a consistent direction when timing is correct, while other signals are reduced according to their cross-correlation with the selected code. If the codes are not orthogonal, the remaining cross-correlation contributes interference. This mathematical representation helps explain why code design and received power both affect the receiver decision.

Simulation Design and Validation

The chip rate determines how quickly the spread waveform changes. If the information bit rate is fixed and the spreading length increases, the chip rate also increases. A higher chip rate requires sufficient signal bandwidth and faster processing in the transmitter and receiver. The implementation must therefore balance processing gain against occupied bandwidth, device power and receiver complexity. It is incorrect to conclude that a very long code can be

selected without consequence. The chosen code length must fit the available bandwidth and the timing capabilities of the system.

A BER experiment should use enough transmitted bits to make the result meaningful. If only 100 bits are tested and no errors occur, the measured BER is zero for that short sample, but this does not prove that the true error probability is zero. Longer runs provide more reliable estimates, especially when the target BER is small. The experiment should report the number of tested bits along with the error count and measured BER. Confidence intervals can be used to show uncertainty when the observed error count is low.

Simulation Design and Validation

The near–far test should specify the power ratio between the strongest and weakest users. A ten-times power difference corresponds to 10 dB, because 10 log10(10) equals 10 dB. The simulation can increase the ratio through 0 dB, 5 dB, 10 dB and 15 dB and record the BER of each user separately. It is important not to report only the average BER because a strong user may perform well while a weak user experiences many errors. Per-user results reveal whether the system is fair and whether power control is effective.

Power control can be evaluated by adding a target received power and a maximum adjustment step. The base station estimates the received signal level and sends a command to increase or decrease the mobile's transmit power. If the feedback delay is too long, the mobile may not respond quickly to fading or movement. If the adjustment step is too large, power can oscillate around the target. A realistic simulation should therefore include measurement error, feedback delay and maximum transmit power. The result should show both BER improvement and the power-control overhead.

Simulation Design and Validation

Timing synchronisation is essential for code correlation. If a mobile's chip sequence arrives late relative to the receiver's local code, the correlation peak may be reduced and cross-correlation with other users may increase. A simple experiment can introduce offsets of zero, one-quarter chip, one-half chip and one chip, then compare correlation values and BER. Multipath can create several delayed versions of the same signal, so the receiver may need a rake-style combining method or an equivalent channel-aware detector. The ideal aligned-code calculation should be treated as a baseline case.

The capacity experiment should define the active-user admission rule. The network may admit a new user only when predicted interference remains below a threshold, or it may accept the user and monitor the measured BER. These approaches can produce different results. An admission-control method protects existing users by refusing a connection before quality degrades too far. A measurement-only method may reveal the physical limit but can allow temporary service degradation. The assignment should explain which policy is being tested and how blocked requests are counted.

Interference and Signal Quality

Code reuse and code assignment should also be considered. If two nearby cells use the same code set, signals from users in neighbouring cells may contribute interference at the receiver. A practical system coordinates code allocation and may use scrambling codes or cell-specific identifiers to distinguish transmissions. The simulator can begin with a single-cell model, then

add one neighbouring cell with a specified path loss and reuse pattern. This extension helps identify how a result from an isolated receiver changes when the network is expanded.
The receiver decision threshold should be applied consistently across all tests. For bipolar data, a positive despread decision statistic represents one symbol and a negative value represents the other. A value near zero is more vulnerable to noise and interference. The model should specify how exact zero values are handled and whether soft information is retained for later decoding. If channel coding is included, report uncoded and coded BER separately because coding gain may conceal the effect of the spreading code itself.

Simulation Design and Validation

A strong CDMA result should include both theoretical expectations and measured evidence. Theoretical processing gain predicts the improvement associated with spreading under simplified assumptions. Simulation reveals how code correlation, power imbalance, timing error and noise affect the actual receiver. The final capacity claim should be limited to the tested conditions and should state the maximum active-user count that met the selected BER target. This makes the conclusion scientifically defensible and avoids presenting a single ideal calculation as proof of unlimited simultaneous access.
A code family should be assessed using its complete correlation matrix. Each diagonal element represents the autocorrelation of a code with itself, while off-diagonal elements represent cross-correlation between different codes. The desired code should have a strong autocorrelation peak and low cross-correlation with other active codes over the relevant timing offsets. Reporting only one orthogonal pair is insufficient when the network needs many simultaneous users. A matrix or maximum cross-correlation value provides a clearer description of the code set's suitability.

Simulation Design and Validation

The noise model should be stated explicitly. Additive white Gaussian noise is a common starting point because it provides a repeatable baseline and allows signal-to-noise ratio to be controlled. A later experiment can add fading, shadowing and multipath. The receiver should use the same noise realisation for corresponding strategy comparisons when possible. The report should specify whether SNR is measured per bit, per chip or over the received signal bandwidth, because these definitions affect the numerical result and can otherwise lead to misleading comparisons.
The receiver can use hard decisions or soft decisions. A hard decision converts the despread statistic immediately into a zero or one, while a soft decision retains information about the strength and reliability of the statistic. Soft information can improve the performance of channel decoding, but it requires additional processing and a suitable decoder. If the assignment focuses only on spreading and correlation, hard decisions are sufficient for the basic experiment. If coding is added, the report should state the decoder and show how coding changes the measured BER.

Simulation Design and Validation

A fair user-capacity test should define the distribution of user powers and positions. If every user is assigned identical received power, the near–far problem is hidden. If one user is always much stronger, the simulation may exaggerate the effect. A useful test can draw path losses

from a stated range and then compare uncontrolled power with a power-control loop. The report should include the distribution of received powers before and after control, not only the average value. This helps explain whether the improvement is due to better power balance or another model change.

The CDMA system should also use admission control when the measured interference becomes too high. A new user can be admitted if the predicted increase in interference leaves all active users above the quality target. If the predicted quality would fall below the threshold, the network can defer or reject the request. This approach protects existing sessions and makes capacity depend on a service-quality constraint rather than on the number of available codes alone. The simulator should count both admitted users and blocked requests so that the capacity result is transparent.

Simulation Design and Validation

The experiment should include a baseline with one active user. This verifies that the transmitter, spreading operation, channel model and receiver are functioning before multiuser interference is introduced. If the single-user BER is already poor, the cause is likely noise level, mapping, synchronisation or receiver implementation rather than code cross-correlation. After the baseline passes, add users gradually and compare the change in BER. This staged approach makes debugging easier and provides a clear reference for interpreting multiuser performance.

A power-control result should report the residual power error after adjustment. If the target received power is Ptarget and the measured power is Preceived, the error in decibels is 10 log10(Ptarget/Preceived). A smaller absolute error indicates that the received powers are more closely balanced. The report should also show how many control updates were needed and whether the mobile reached its maximum transmit power. A system that achieves a low BER only by requiring impractically high transmit power may not be suitable for battery-powered devices.

Capacity and Resource Utilisation

The final capacity figure should include a margin below the point where BER first exceeds the target. Radio conditions vary over time, so operating exactly at the measured limit can cause unstable service. A practical admission threshold can be set below the maximum tested user count to provide room for fading and unexpected interference. The margin should be justified using repeated trials and the variation in BER. This converts a laboratory result into a more cautious network-planning recommendation.

The results should be presented separately for users with different received powers. A network-wide average BER can hide a weak user whose error rate is unacceptable. The report should include the best, median and worst user BER, together with the received-power spread and code cross-correlation. If power control improves the worst-user result while slightly increasing total power, that trade-off should be stated. The purpose of the analysis is to demonstrate reliable multiuser operation, not simply to maximise the number of active codes.

Simulation Design and Validation

The CDMA recommendation should remain limited to the model conditions that were tested. If the simulation assumes perfect timing, equal power and ideal channel estimation, the

conclusion should not claim performance under arbitrary mobility or interference. Further tests should add timing error, multipath, power-control delay and neighbouring-cell interference. Stating these limitations does not weaken the work; it shows which assumptions support the result and what evidence is still required for a practical network.

ANSWER – Q4: OFDMA Subcarrier Allocation

1. Subcarrier spacing and symbol duration

For an OFDM system with N equally spaced orthogonal subcarriers, $\Delta f = 1/Tu$, where Δf is subcarrier spacing and Tu is the useful symbol duration. For example, if $\Delta f = 15$ kHz, $Tu = 1/15{,}000 = 66.67$ μs.

A cyclic prefix (CP) copies the end of the OFDM symbol to its beginning. It reduces inter-symbol interference caused by multipath delay, provided the CP is longer than the relevant channel delay spread. It adds overhead, so the total symbol duration is $Ts = Tu + Tcp$.

2. Numerical orthogonality verification

For subcarriers k and m over the useful symbol interval, the inner product is proportional to $\int_0^{Tu} \exp(j2\pi(k-m)t/Tu)dt$. If $k \neq m$ and the interval is exactly Tu, the positive and negative rotations cancel and the integral is zero. For $k = m$, the integral equals Tu. For example, with $Tu = 1$ ms and $\Delta f = 1$ kHz, subcarriers at 0, 1 kHz and 2 kHz complete integer cycles during the interval; their pairwise normalised inner products are ideally 0, 0 and 0.

Pair	**Frequency difference**	**Cycles across Tu**	**Ideal normalised inner product**
k=0, m=1	1 kHz	1	0
k=0, m=2	2 kHz	2	0
k=1, m=2	1 kHz	1	0

3. Allocation approaches

Approach	**Rule**	**Expected benefit**	**Risk**
Fixed contiguous blocks	Give each user 8 adjacent subcarriers (64 ÷ 8)	Simple and predictable; each user gets equal resource count	A user may receive a block with poor fading gains.
Interleaved round-robin	Allocate every eighth subcarrier to each user	Spreads each user's resources across frequency	Does not necessarily assign the best channel to the user.

Channel-aware best-subcarrier	Assign each subcarrier to the user with the highest channel quality	Can maximise sum throughput	Weak users may receive few resources and fairness can fall.

4. Step-by-step throughput and fairness method

Generate a 8 × 64 channel-gain matrix for each channel realisation using a fixed seed.

For each allocation, compute each assigned subcarrier rate using $r = \log2(1 + SNR \times gain)$ bits/s/Hz (or the same chosen modulation table for all schemes).

Sum assigned rates to obtain total throughput and sum each user's assigned rates to obtain per-user throughput.

Compute Jain's fairness index: $J = (\Sigma xi)^2 / (n \times \Sigma xi^2)$, where xi is user i's throughput and n = 8. J = 1 means perfectly equal rates.

5. Illustrative result table

The following values are a worked example to show how to report the results; actual values must be produced by running the fixed-seed simulation over at least 1,000 realisations.

Allocation	**Example sum throughput (relative units)**	**Example Jain fairness**	**Interpretation**
Fixed contiguous	42	0.96	Fair and predictable, but may miss good subcarriers.
Interleaved round-robin	46	0.98	Frequency diversity improves robustness and equal resource counts help fairness.
Channel-aware best-subcarrier	58	0.62	Highest sum throughput, but strong-channel users can dominate.

6. Design decision

Data Synchronisation and Consistency

For a lecture hall, use proportional-fair or minimum-rate-constrained channel-aware allocation. First reserve enough subcarriers to meet each student's minimum rate, then distribute remaining subcarriers according to channel quality. This balances throughput and fairness. Pure maximum-throughput allocation is unsuitable if it repeatedly starves weak users.

OFDMA divides the available channel bandwidth into many narrow, mutually orthogonal subcarriers. The scheduler assigns groups of subcarriers to users according to traffic demand, channel quality and fairness requirements. Orthogonality allows adjacent subcarriers to overlap spectrally without causing ideal inter-carrier interference when the transmitter and receiver maintain the required timing and frequency accuracy. In practice, frequency offset, phase noise, Doppler spread and timing errors can reduce orthogonality. The cyclic prefix and synchronisation procedures help manage multipath and timing effects, but they introduce overhead and do not remove every source of impairment.

Interference and Signal Quality

The relationship between subcarrier spacing and useful symbol duration is $\Delta f = 1/Tu$. With a spacing of 15 kHz, the useful symbol duration is approximately 66.67 microseconds. A smaller spacing gives a longer useful symbol and may be more sensitive to time variation, while a larger spacing gives a shorter symbol and can be more suitable for higher mobility. The exact numerology selected in a real system depends on the standard, carrier frequency, delay spread, mobility and deployment type. The numerical example in the assignment is intended to demonstrate the reciprocal relationship rather than prescribe one universal configuration.

The cyclic prefix is formed by copying the end portion of an OFDM symbol and placing it at the beginning. If the prefix is longer than the significant channel delay spread, delayed copies from the previous symbol can be absorbed within the prefix interval and the useful symbol can be processed with reduced inter-symbol interference. The prefix consumes time that does not carry new information, so a longer prefix reduces spectral efficiency. The system designer must balance multipath protection against overhead. The prefix should be chosen using channel measurements and the expected deployment environment.

Capacity and Resource Utilisation

Orthogonality can be verified by calculating the inner product of two subcarrier waveforms over the useful symbol duration. When their frequency difference is an integer multiple of $1/Tu$, the complex exponential completes an integer number of cycles and the integral is zero for distinct subcarriers. The desired subcarrier has a non-zero self-inner-product. This mathematical result explains why the subcarriers can overlap in frequency while remaining separable at the receiver. A numerical implementation should sample the waveforms at a sufficiently high rate and check that the normalised inner products are close to zero within numerical precision.

Fixed contiguous allocation gives each user a continuous block of neighbouring subcarriers. It is simple to schedule and can be useful when users have similar channel conditions or when contiguous allocation is required by the system. However, a user may receive subcarriers that

experience deep fading, while another part of the band has better channel quality. The method also makes it harder to exploit frequency diversity when the channel varies significantly across the bandwidth. Its main advantage is predictable allocation and relatively straightforward resource management.

Capacity and Resource Utilisation

Interleaved round-robin allocation distributes each user's subcarriers across the frequency band. This can provide frequency diversity because one user's resources are less likely to be affected by a single narrow frequency fade. Equal numbers of subcarriers also make the resource count easy to compare. However, equal resource counts do not guarantee equal data rates because users may experience different signal-to-noise ratios and channel gains. A fairness assessment should therefore compare both allocated resource units and achieved throughput. The scheduler may need to compensate users with poor channel conditions if minimum service guarantees are required.

Channel-aware allocation assigns each subcarrier to the user with the best measured channel quality or the highest expected rate. This can increase total system throughput because resources are assigned where they can carry more bits. The disadvantage is that users with consistently weak channels may receive very few resources. A pure maximum-throughput policy can therefore produce unfair service even when aggregate throughput is high. Proportional-fair scheduling addresses this by comparing current achievable rate with a user's historical average rate. Minimum-rate constraints can also ensure that every user receives a basic level of service.

Simulation Design and Validation

The throughput calculation should use the same channel-gain matrix, noise level and modulation assumptions for all allocation methods. For each assigned subcarrier, calculate the rate using the selected link model, then sum rates for each user and for the entire system. Jain's fairness index provides a compact measure of how evenly throughput is distributed. A value near one indicates similar user throughputs, while a lower value indicates greater inequality. Fairness should not be interpreted in isolation: a perfectly fair system can still deliver poor total throughput, and a high-throughput system can still starve some users.

A reliable experiment should generate channel gains for several independent realisations and use a fixed seed for repeatability. One realisation may favour one allocation method by chance, especially when only a small number of users and subcarriers are simulated. Repeating the test over at least 1,000 channel realisations allows the mean and variation of throughput and fairness to be estimated. Report confidence intervals or standard deviations where appropriate. Also record the number of subcarriers assigned to each user, the number of users below the minimum rate and the scheduling time required to make an allocation decision.

User Interface and Network Status

For the lecture hall, the most balanced approach is a channel-aware scheduler with proportional fairness or a minimum-rate constraint. The scheduler should first ensure that each user receives sufficient resources to meet the required minimum service, then use remaining subcarriers to improve aggregate throughput. The design should be tested under different user

counts, channel conditions and traffic loads. The final selection should explain the trade-off between sum rate and fairness, rather than choosing the method with the largest illustrative throughput value without considering whether all users receive acceptable service.

The OFDMA scheduler operates over a time-frequency resource grid. The frequency axis contains subcarriers or groups of subcarriers, while the time axis contains OFDM symbols or scheduling intervals. A resource block is a defined group of subcarriers over one or more symbols. The scheduler assigns these resources according to channel state information, queue demand and service priority. The quality of the allocation depends on how frequently channel information is updated. If feedback is delayed, the scheduler may assign a user resources based on channel conditions that have already changed.

User Interface and Network Status

Channel state information can be estimated from pilot or reference signals and reported to the scheduler. The report may include channel quality, rank or preferred precoding information depending on the system. Frequent feedback improves scheduling accuracy but consumes uplink resources and device energy. In a high-mobility environment, channel conditions can change quickly, making stale feedback less useful. The simulator should therefore test perfect channel knowledge and delayed or noisy channel estimates separately. This comparison shows how much of the channel-aware throughput gain depends on accurate feedback.

The Shannon-style rate expression $r = \log2(1 + SNR \times gain)$ provides an idealised upper-bound relationship between signal quality and spectral efficiency. A real system uses discrete modulation and coding schemes, so the rate changes in steps rather than continuously. Coding overhead, control signals, reference symbols and retransmissions reduce the useful payload rate. If the assignment uses the logarithmic expression, it should state that the result is a theoretical model and apply the same expression to every allocation method. A separate practical model can use a fixed modulation-and-coding table.

Capacity and Resource Utilisation

Frequency-selective fading means that different subcarriers can experience different channel gains. A contiguous block may place a user on several adjacent subcarriers that all lie in a faded part of the spectrum. An interleaved allocation can distribute resources across frequency and reduce the risk that all assigned subcarriers experience the same fade. Channel-aware allocation can select the strongest subcarriers for each user, but the scheduler must also account for fairness and signalling overhead. These differences explain why allocation strategy matters even when the total number of subcarriers is unchanged.

The cyclic prefix should be included when calculating net throughput. If the useful symbol duration is Tu and the prefix duration is Tcp, the fraction of symbol time carrying useful data is Tu divided by Tu plus Tcp, before other overheads are considered. A longer prefix provides more protection against delay spread but lowers the net rate. A short prefix improves efficiency but may fail in environments with long multipath delays. The design should use a propagation model that reflects the lecture hall or other deployment rather than assuming that every radio channel has the same delay spread.

Simulation Design and Validation

Inter-carrier interference can be introduced by adding a frequency offset to the received OFDM waveform. Even a small offset can cause energy from one subcarrier to leak into another because the receiver's frequency reference no longer matches the transmitter. Doppler shift from user movement can create a similar problem. The simulation should compare ideal synchronisation with several offset values and record the resulting signal-to-interference ratio or bit-error rate. This experiment demonstrates that orthogonality depends on accurate timing and frequency synchronisation.

The fairness index should be accompanied by per-user throughput values. Two allocation methods can have the same Jain index but different total throughput and different minimum user rates. A useful table should report aggregate throughput, average throughput, minimum throughput, maximum throughput and fairness. It should also count users who fail to meet a specified service threshold. These measurements help determine whether an allocation method is suitable for a classroom in which every student may need a minimum video or interactive-data rate.

Simulation Design and Validation

A proportional-fair scheduler typically compares the instantaneous achievable rate with a user's historical average throughput. A user with a temporarily strong channel can receive more resources, but a user who has received little service may also receive priority because its historical average is low. The averaging window controls the trade-off between responsiveness and fairness. A short window reacts quickly to changing channels but may create unstable allocations, while a long window produces smoother service but responds more slowly. The simulation should state the averaging parameter and test more than one value.

The 8-user by 64-subcarrier channel-gain matrix should be generated consistently across the three allocation strategies. The matrix can represent independent fading values or a correlated frequency-selective channel. The selected distribution should be documented, including its mean and variance. If the same matrix is reused for each method, differences in output can be attributed to the allocation rule. If a new matrix is generated for each method, random variation may dominate the comparison. Repeating the same experiment over many independent matrices provides a fair and statistically meaningful evaluation.

Data Synchronisation and Consistency

The recommended lecture-hall solution should include admission control when the number of users exceeds the available resource capacity. The scheduler can prioritise essential traffic, enforce a minimum rate for admitted users and reject or defer new sessions when no acceptable allocation is possible. Without admission control, a system may admit every user but deliver an unusably low rate to all of them. The final evaluation should therefore include both throughput and the number of users that receive the required minimum service. This gives a more realistic measure of system quality than aggregate rate alone.

The scheduler must account for control resources that are not available for user payload. Reference signals, synchronisation signals, broadcast information and scheduling assignments consume part of the time-frequency grid. If these resources are omitted, the calculated throughput will be higher than the practical payload throughput. A consistent comparison

should reserve the same required control overhead for each allocation strategy or explain why the overhead differs. The result should clearly identify whether it represents gross physical-layer capacity or net user data rate.

Simulation Design and Validation

The channel-aware algorithm needs a rule for handling users whose channel reports are missing or outdated. It can use a conservative average channel estimate, assign a minimum resource share or temporarily schedule the user using a robust modulation and coding level. Without such a rule, a user with missing feedback may be starved even though the application still has data to send. The simulation should include feedback loss or delay and record how often users are scheduled using fallback information. This gives a more realistic view of scheduler reliability.

A minimum-rate constraint can be implemented by first identifying the resources required for each user to reach the target rate. If the total required resources exceed the available grid, the system cannot meet every user's target simultaneously. It must then apply admission control, reduce the target or allocate resources according to priority. This feasibility check should occur before the scheduler distributes remaining subcarriers to maximise throughput. The approach prevents a high aggregate rate from hiding the fact that some users receive no acceptable service.

Simulation Design and Validation

The scheduler's computational complexity also matters. A simple round-robin allocator can make decisions quickly, while a channel-aware optimiser may need to evaluate many user-subcarrier combinations. In a large system, the decision must be completed within the scheduling interval. The experiment should record execution time as a supporting measure, but it should also count the number of candidate assignments evaluated and the quality of the resulting allocation. A slow optimiser may be unsuitable for a short scheduling interval even if it produces a slightly higher theoretical throughput.

The final OFDMA result should include a clear statement of model limitations. An ideal logarithmic rate equation, perfect channel knowledge and perfect synchronisation are useful for learning but do not represent every practical radio condition. A more complete model can add modulation limits, coding overhead, cyclic-prefix loss, control resources, channel-estimation error and retransmissions. The conclusion should distinguish results from the simplified model from claims about a deployed network. This keeps the analysis accurate and makes the assumptions available for future improvement.

Simulation Design and Validation

The allocation method should be tested with both symmetric and asymmetric traffic demand. In a symmetric case, every user has a similar amount of data to send. In an asymmetric case, one or two users may have much larger queues than the others. Round-robin allocation may waste resources on users with little data, while a purely channel-aware method may favour users with strong signals. A proportional-fair scheduler can balance queue demand, channel quality and historical service. The experiment should report how each method behaves as the demand distribution changes.

A practical OFDMA system also uses retransmission mechanisms when a packet is decoded incorrectly. Retransmissions consume additional resources and reduce the net throughput available to new data. The scheduler may allocate a more robust modulation and coding scheme to users with poor channel quality, increasing reliability but reducing bits per resource unit. The simulation should distinguish first-transmission throughput from successfully delivered payload throughput. This ensures that a high physical-layer rate is not mistaken for high application goodput.

User Interface and Network Status

The selected allocation should be reviewed when the number of users exceeds the scheduler's design range. More users create more candidate assignments and may increase feedback overhead. The scheduler can group users, limit feedback frequency or use a simpler allocation rule when the system is heavily loaded. Such measures reduce computation and signalling but may reduce throughput or fairness. A scalability test should therefore record both radio performance and the time required to calculate an allocation.

The final throughput table should identify whether values are expressed in bits per second, bits per second per hertz or normalised rate units. These quantities cannot be compared without the bandwidth and time interval used in the calculation. The table should also distinguish gross rate from net payload rate after control and retransmission overhead. Clear units make the results reproducible and prevent an illustrative relative-throughput value from being mistaken for a measured data rate.

Capacity and Resource Utilisation

The OFDMA design should be tested with different channel conditions, including flat fading and frequency-selective fading. Flat fading applies approximately the same gain across the allocated bandwidth, while frequency-selective fading creates stronger and weaker subcarrier regions. Channel-aware allocation is expected to show a greater benefit when frequency variation is present. Reporting both conditions demonstrates whether the scheduler's improvement comes from exploiting frequency diversity or from another feature of the model.

ANSWER – Q5: SDMA and Directional Antennas

1. Uniform linear array (ULA) array factor

For N equally spaced antenna elements with spacing d, wavelength λ, and steering angle θ_0, a normalised array factor can be written as $AF(\theta) = |\sin(N\psi/2) / (N \sin(\psi/2))|$, where $\psi = (2\pi d/\lambda)(\sin\theta - \sin\theta_0)$. At $\psi = 0$, the limiting value is 1 (the main-beam maximum).

2. Prediction before calculation

As N increases, the main lobe becomes narrower and angular discrimination improves.

The ideal uniform-amplitude array's first sidelobe is approximately −13 dB relative to the main lobe; increasing N narrows the beam but does not automatically eliminate sidelobes.

Closely spaced users have similar steering angles, so the beam may not separate them well.

An omni antenna provides little angular discrimination; a 120° sector antenna provides coarse separation; a steered array can provide finer separation.

3. Worked numerical setup

Assume half-wavelength spacing $d = \lambda/2$ and steer the array to user A at $\theta_0 = 0°$. Then $\psi = \pi \sin\theta$. For a user B at angular separation θ, the normalised response is $AF(\theta) = |\sin(N\pi \sin\theta/2) / (N \sin(\pi \sin\theta/2))|$. Approximate power response is $|AF|^2$.

To evaluate interference, define SIR threshold = 10 dB. If user A's transmitted power is normalised to 1 and the response toward B is $|AF(\theta)|^2$, a simple single-interferer illustration is SIR_B ≈ desired power / leakage power. A full multiuser SIR also requires the desired user's own beam gain, other-user powers, and noise.

Array elements	**Approx. broadside HPBW**	**Ideal first sidelobe**	**Expected angular separation**
N = 2	≈ 60°	≈ −13 dB	Largest; broad beam has weak separation
N = 4	≈ 29°	≈ −13 dB	Better than 2 elements
N = 8	≈ 14°	≈ −13 dB	Can separate more closely spaced users
N = 16	≈ 7°	≈ −13 dB	Narrowest main beam among the cases

The beamwidth figures are approximate broadside estimates, not exact simulation output. Exact HPBW, sidelobe level, and minimum separation must be computed from sampled patterns for 5°, 10°, …, 60° and checked against the 10 dB threshold. Record the first separation where the measured SIR reaches or exceeds 10 dB.

4. Why SDMA has limits

SDMA is limited by angular spread, multipath, array calibration, finite antenna count, mutual coupling, and users clustered in the same direction. Two users nearly aligned in angle produce similar spatial signatures; the receiver cannot reliably distinguish them using direction alone.

5. Toll-plaza decision

At a toll plaza, most vehicles approach from nearly the same direction, so relying only on SDMA is risky. Use sector coverage and time/frequency scheduling, with SDMA as an additional gain when vehicles are sufficiently separated in angle. The hybrid is more robust than a beam-only solution.

Space-division multiple access separates users by their spatial signatures rather than assigning every user a completely separate frequency or time resource. A base station with multiple antenna elements can shape its transmitted or received beam toward a desired direction. If two users have sufficiently different angles of arrival, the antenna array may distinguish them and reuse the same time-frequency resource. The achievable separation depends on array size,

element spacing, wavelength, calibration, propagation and the angular distance between users. SDMA is therefore a way to improve resource reuse, not a guarantee that users can always be separated by direction.

Interference and Signal Quality

The uniform linear array model assumes equally spaced antenna elements arranged along a straight line. Each element receives a phase-shifted version of a plane wave, and the array combines these signals using selected weights. The array factor describes the directional response of the array. Steering changes the phase progression so that signals from a chosen direction add constructively. The model is useful for explaining beam formation, but a real antenna system also has element patterns, mutual coupling, feed losses, manufacturing tolerances and environmental reflections. These effects should be included when moving from a theoretical pattern to an installed system.

With half-wavelength spacing, the phase difference between adjacent elements changes with the sine of the arrival angle. The steering weights compensate for the phase progression at the desired angle. At the steering direction, the normalised array factor reaches its maximum value. At other directions, partial cancellation reduces the response, producing the main lobe, sidelobes and nulls. The exact pattern depends on the number of elements and the weighting applied. Uniform weighting is simple, while tapering can reduce sidelobes at the cost of a wider main lobe or reduced peak gain.

Interference and Signal Quality

Increasing the number of antenna elements generally narrows the main beam and improves angular discrimination. It does not automatically eliminate sidelobes or make two very closely spaced users separable. The ideal first sidelobe level for a uniformly weighted array is approximately minus 13 dB, but practical patterns can differ. A larger array also requires more radio-frequency chains or more complex beamforming hardware, depending on the architecture. The additional hardware increases cost, power consumption, calibration requirements and signal-processing load. The design should therefore choose array size based on the required separation and deployment budget.

The half-power beamwidth is the angular width between points where the power response falls to half of its maximum value. It is a useful summary of main-lobe width, but it is not sufficient by itself to determine whether two users can be separated. The sidelobe level and null locations also influence interference. A numerical test should sample the full angular pattern, convert the magnitude response to power, and calculate the response toward interfering users. The minimum separation should be defined using a stated SIR threshold and a multiuser interference model rather than inferred from beamwidth alone.

Interference and Signal Quality

A single-interferer illustration can help explain directional leakage, but it should not be confused with a complete multiuser SIR calculation. A full model includes desired-signal gain, the transmit power of every interfering user, each user's angle, antenna response toward the receiver, propagation loss and receiver noise. The total interference power is the sum of the individual contributions in linear units. Only after summing the powers should the result

be converted to decibels. Adding interference values directly in dB would produce an incorrect result.

SDMA performance is affected by multipath because reflected signals can arrive from directions different from the direct path. A receiver may observe several spatial components for one user, and those components can overlap with another user's spatial signature. Users located in nearly the same direction may be difficult to separate even if they are physically far apart. Conversely, users at similar distances may be separable if their angles are sufficiently different. This is why spatial separation should be evaluated using measured or simulated channel vectors rather than only a map of user positions.

Capacity and Resource Utilisation

Array calibration is also important. Small phase or amplitude errors between antenna elements can shift the main beam, raise sidelobes or reduce null depth. Mutual coupling changes the effective response of each element, while nearby structures can distort the radiation pattern. The system should be calibrated after installation and periodically checked. A practical test can compare the measured pattern with the predicted pattern at several steering angles. Any difference should be documented and used to update the beamforming weights or the coverage plan.

At a toll plaza, vehicles approach through lanes and may have similar directions of travel. A beamforming system may separate vehicles in different lanes under favourable geometry, but it should not rely on angle alone because vehicles can be closely aligned. Sector coverage, time/frequency scheduling and SDMA can be combined. Scheduling provides a dependable baseline, while spatial processing reuses resources when channel conditions allow. The controller should avoid assigning the same resource to users whose spatial signatures are too similar or whose measured SIR falls below the selected threshold.

Capacity and Resource Utilisation

The deployment decision should be based on measured user separation, required throughput, antenna cost and operational complexity. A small array may be sufficient when users are spread across distinct directions, while a larger array may be justified in a dense environment with high demand. The design should report the beam pattern, half-power beamwidth, sidelobe level, minimum tested angular separation, SIR distribution and outage probability. These values allow the SDMA option to be compared fairly with time- or frequency-based alternatives. The conclusion should state the conditions under which spatial reuse provides a benefit and where a hybrid method is safer.

The wavelength is related to carrier frequency by the speed of light divided by frequency. As frequency increases, wavelength decreases, so a half-wavelength antenna spacing becomes physically smaller. The same number of elements can therefore fit into a smaller array at a higher carrier frequency. However, higher frequencies may experience greater propagation loss and blockage, and the antenna design becomes more sensitive to manufacturing tolerances. Array size should be selected with both electromagnetic behaviour and physical installation constraints in mind.

Capacity and Resource Utilisation

The element spacing should not be increased without limit. If elements are spaced too far apart, grating lobes can appear and create unwanted high-gain directions. These lobes may cause interference or allow the array to receive signals from directions that were intended to be suppressed. Half-wavelength spacing is a common reference because it helps avoid grating lobes over a broad steering range in a simple linear-array model. A practical design should verify the full steering range and include the actual element pattern.

Beam steering can be performed using phase shifts, digital weights or a hybrid combination of analogue and digital processing. Analogue beamforming may use fewer radio-frequency chains and lower hardware cost, but it offers less flexibility when serving multiple users simultaneously. Digital beamforming can form several beams and apply more advanced spatial processing, but it requires more conversion and processing resources. The appropriate architecture depends on the number of simultaneous streams, bandwidth, power budget and cost target. The array-factor equation alone does not capture these implementation differences.

Interference and Signal Quality

Antenna sidelobes are important because they determine how much energy is directed away from the main beam. A user located in a sidelobe direction may still receive a significant interfering signal. Tapering the array weights can reduce sidelobe levels, but it usually broadens the main lobe and may reduce gain. The trade-off should be evaluated using the complete radiation pattern rather than a single sidelobe figure. The simulator should record the main-lobe gain, half-power beamwidth, first sidelobe level and response at the angles occupied by interfering users.

The SIR calculation should sum all relevant interferers in linear power units. If the desired received power is Pd and interfering powers are P1 through Pm, the interference-limited ratio is Pd divided by the sum of those powers, with noise added to the denominator when the system is noise-limited. Converting each power to dB before summing would be incorrect. The report should show at least one worked numerical example in linear units and then convert the final ratio to decibels using 10 log10(SIR).

Simulation Design and Validation

User distribution has a strong effect on spatial reuse. If users are uniformly spread across a service area, their arrival angles may be diverse and SDMA can separate many of them. If users are concentrated along a road or in a narrow corridor, their angles may be similar and spatial separation becomes difficult. A useful simulation should compare uniform angular distribution, clustered distribution and a single-direction approach such as a toll-plaza lane. The results should report the number of users that can share the same resource while meeting the chosen SIR target.

The array model should include receiver noise as well as interference. In a high-interference environment, the SIR may be the dominant measure. In a low-interference environment, thermal noise can limit the signal-to-interference-plus-noise ratio. The received signal power also depends on path loss, shadowing and antenna gain. A complete link budget should identify transmit power, antenna gains, feeder losses, propagation loss and receiver noise figure. This provides a consistent basis for comparing users at different distances and directions.

Simulation Design and Validation

Mobility can change a user's angle and channel response over time. A beam that initially points toward a user may become misaligned as the user moves, particularly if the beam is narrow. The system must update beam weights and hand over the user between sectors or sites when necessary. The update rate should be fast enough to track movement but not so fast that signalling overhead becomes excessive. The experiment can introduce angular movement and measure beam-tracking error, outage duration and the number of beam updates.

The toll-plaza hybrid should use sector coverage to provide robust baseline service and apply SDMA when the measured spatial signatures are sufficiently distinct. A scheduler can assign different time or frequency resources to vehicles that approach from nearly the same direction. This avoids relying on an ideal beam pattern to separate users that are physically aligned. The system can also prioritise vehicles carrying urgent transaction data so that congestion does not cause long delays. The design should be evaluated during both normal traffic and a peak vehicle-arrival period.

Capacity and Resource Utilisation

The final comparison should state the hardware and operational costs of the array. More elements can improve directivity, but they may require additional amplifiers, phase shifters, converters, calibration and processing. Power consumption and thermal management can become important at a roadside installation. The design should also consider maintenance access and weather protection. A larger array is justified only if the measured improvement in capacity, SIR or coverage is sufficient to offset these costs. This makes the recommendation a complete engineering decision rather than a pattern-only calculation.

The spatial channel of a user can be represented by a vector containing the complex response at each antenna element. Users with sufficiently different channel vectors can be separated using beamforming or multiuser detection. Two users at different physical angles may still have correlated channel vectors in a rich multipath environment, while users at similar angles may sometimes be separated by differences in their propagation paths. This is why the spatial channel vector is a stronger basis for SDMA evaluation than angle alone.

Simulation Design and Validation

The array should be evaluated over the full service sector rather than only at broadside. The beamwidth and sidelobe pattern can change as the steering direction moves away from the centre of the array's field of view. If the system must serve users over a wide angular range, the simulation should calculate the pattern at multiple steering angles. It should identify directions where gain falls or sidelobes rise. These results help determine whether one array can cover the full area or whether several sectors are needed.

The number of simultaneous spatial streams is limited by the number of effective antenna dimensions and by channel correlation. Adding antenna elements does not guarantee an equal increase in user capacity because the channel may not provide enough independent spatial paths. If users are clustered or the environment has strong line-of-sight alignment, the channel matrix may have a low effective rank. A realistic capacity test should calculate channel correlation or singular values and report how many spatial streams can be supported under the selected quality threshold.

Simulation Design and Validation

The beamforming system should include a fallback when a user's spatial channel becomes highly correlated with another active user. The scheduler can move one user to another time or frequency resource, change the beam weights or reduce the number of simultaneous streams. This avoids maintaining an allocation that looks efficient in the resource table but fails to provide adequate SIR. The fallback action should be logged so that the operator can identify how often spatial reuse is possible and how often the system must revert to orthogonal scheduling.

The toll-plaza test should include lane geometry, vehicle speed and vehicle density. As vehicles move through the coverage area, their angles and distances change, and the system may need to update beam weights. A high vehicle density can also create temporary blockage and reflections from nearby vehicles or structures. The experiment should record the duration of each successful spatial allocation, the number of beam updates and any service interruption during a transition. This gives a more realistic measure of the benefit of SDMA in a moving-vehicle environment.

Capacity and Resource Utilisation

The antenna array should be installed with a clear view of its intended service area where possible. Nearby walls, vehicles, poles and building surfaces can reflect or block radio waves and change the pattern observed in practice. A theoretical free-space array factor is therefore a starting point, not a complete coverage prediction. A site survey can identify major obstructions and reflection paths. After installation, drive tests or fixed-point measurements should be compared with the predicted coverage to identify any adjustment required.

A beamforming system must also protect against abrupt changes in the active-user set. When a vehicle leaves the coverage area, its resource should be released and the scheduler should update the beam configuration for remaining users. When a new vehicle enters, it should be admitted only after the system checks available spatial and time-frequency resources. This avoids keeping stale beam assignments and reduces unnecessary interference. The controller should log admission, release and beam-update events for later analysis.

Simulation Design and Validation

The final recommendation should state whether the expected gain comes from better antenna gain, reduced interference or simultaneous spatial reuse. These are different effects. Antenna gain can improve the link budget for one user, while spatial reuse increases the number of users sharing the same resource. A simulation should report them separately so that the benefit is not overstated. The cost of additional elements and signal processing should then be compared with the measured improvement in user capacity and service quality.

The array-factor model should state whether angles are measured from broadside or from the array axis. A change in angle convention changes the expression used for phase difference and can lead to incorrect steering results. The simulation should use one convention consistently in the equations, plots and tables. It should also state the angular sampling interval so that narrow peaks and nulls are not missed between sample points.

Data Synchronisation and Consistency

The SDMA conclusion should include a practical fallback for users whose spatial channels cannot be separated. These users can be assigned different time or frequency resources, or the scheduler can delay one connection until a resource becomes available. This hybrid rule protects service quality when user geometry is unfavourable. It also demonstrates that SDMA is most useful as an additional reuse opportunity rather than as the only access method in every environment.

ANSWER – Q6: Selecting Multiple-Access Techniques

1. Weighted decision criteria

Rate each method from 1 (poor) to 5 (excellent). Weights total 100%. The weights are design assumptions for this assignment and should be adjusted if the deployment priorities change.

Criterion	**Weight**	**Reason**
Spectral efficiency	25%	Important when bandwidth is limited.
Latency / real-time support	20%	Important for voice and interactive services.
Robustness to interference	20%	Important in tactical and crowded radio environments.
Scalability / many devices	20%	Important for stadiums and 5,000 sensors.
Fairness / predictable service	15%	Avoids starving users and supports service guarantees.

2. Rating matrix (1 = poor, 5 = excellent)

Technique	**Spectral efficiency**	**Latency**	**Interference robustness**	**Scalability**	**Fairness**
SDMA	4	4	3	4	3
TDMA	3	4	3	3	4
FDMA	2	4	3	2	4
CDMA	4	3	4	4	3
OFDMA	5	4	3	5	4

3. Weighted-score calculation example

For OFDMA: score = 0.25(5) + 0.20(4) + 0.20(3) + 0.20(5) + 0.15(4) = 1.25 + 0.80 + 0.60 + 1.00 + 0.60 = 4.25 out of 5.

Technique	**Weighted score / 5**
SDMA	0.25(4)+0.20(4)+0.20(3)+0.20(4)+0.15(3) = 3.65
TDMA	0.25(3)+0.20(4)+0.20(3)+0.20(3)+0.15(4) = 3.35
FDMA	0.25(2)+0.20(4)+0.20(3)+0.20(2)+0.15(4) = 2.90
CDMA	0.25(4)+0.20(3)+0.20(4)+0.20(4)+0.15(3) = 3.65
OFDMA	4.25

4. Choices for four deployments

Deployment	**Primary choice**	**Useful hybrid**	**Reason**
Wide-area cellular voice	TDMA/FDMA-style resource structure or OFDMA in modern systems	FDMA + TDMA / OFDMA + SDMA	Regular resource allocation supports voice quality and frequency reuse.
Jam-resistant tactical link	CDMA / spread-spectrum approach	Frequency hopping and coding where available	Spreading can make interference harder to exploit, although it does not make a link immune to jamming.
High-rate stadium downlink	OFDMA	OFDMA + SDMA/MIMO	Subcarriers can be assigned based on channel quality and spatial streams can reuse resources.

5,000 low-rate sensors	Scheduled/shared access rather than one dedicated channel per device	TDMA scheduling with aggregation and sleep cycles	Small, infrequent reports make permanent dedicated allocation inefficient.

5. Sensitivity check

Change each criterion weight by ±20%, then renormalise weights to 100% and recalculate scores. OFDMA is expected to remain a strong choice for high-rate stadium traffic because it scores highly in spectral efficiency and scalability. The tactical-link decision is less robust: increasing the robustness weight favours CDMA/spread spectrum, while prioritising deterministic latency may favour scheduled TDMA. The sensor decision is also sensitive to device synchronisation, battery life, and traffic burstiness.

Conclusion: Real systems combine methods because no single technique optimises every requirement. FDMA separates frequency bands, TDMA separates time, CDMA separates codes, OFDMA allocates orthogonal subcarriers, and SDMA reuses resources spatially. Hybrid designs balance capacity, latency, interference, and fairness.

Link Availability and Offline Operation

Multiple-access techniques solve the same broad problem of allowing many users to share limited radio resources, but they organise that sharing differently. FDMA separates users by frequency, TDMA by time, CDMA by code, OFDMA by orthogonal subcarriers and SDMA by spatial direction or channel signature. Each method has advantages under particular traffic and propagation conditions. The best technique cannot be selected by comparing one characteristic alone. A practical decision should include spectral efficiency, latency, interference behaviour, scalability, fairness, device complexity, power consumption and the availability of suitable infrastructure.

The weighted decision matrix converts qualitative priorities into a repeatable comparison. Each criterion receives a weight that represents its importance to the deployment, and each technique receives a rating using the same scale. The weighted score is the sum of each rating multiplied by its normalised weight. The result is useful for organising a decision, but it is not an objective physical law. The rating definitions should be documented, and the weights should be agreed before the alternatives are scored. Otherwise, the matrix can be adjusted after the fact to favour a preferred technology.

Data Synchronisation and Consistency

Spectral efficiency measures how much useful information is carried per unit of bandwidth. It is important when spectrum is scarce, but a high spectral-efficiency value does not guarantee low latency or fair access. A system may achieve high aggregate throughput by allocating most resources to users with favourable channels. Latency measures the time between a request or packet arrival and successful delivery. For voice and control applications, predictable delay may be more important than maximum average throughput. The decision

matrix should therefore treat these measures separately rather than combining them into a vague notion of performance.

Interference robustness describes how well a technique maintains service when other transmissions, multipath or external disturbances are present. CDMA uses spreading and correlation to manage multiple simultaneous signals, but it remains sensitive to near–far conditions and code correlation. OFDMA uses orthogonal subcarriers, but frequency offset and synchronisation errors can cause inter-carrier interference. SDMA depends on spatial separation and antenna calibration. FDMA and TDMA can simplify separation in some settings, but they still require frequency planning or accurate timing. Every method has an interference mechanism that must be tested under realistic assumptions.

Data Synchronisation and Consistency

Scalability concerns how the system behaves as the number of users, devices or traffic sources increases. A method that performs well with a few active users may become difficult to manage when thousands of devices generate requests. The system may need more scheduling messages, code management, synchronisation or radio-frequency hardware. Low-rate sensors create a different scaling challenge from high-rate mobile broadband users. Their average data volume may be small, but simultaneous reconnect attempts can produce a large burst. Scalability should therefore be assessed using both average load and peak concurrent demand.

Fairness is important when users share a finite resource. Equal resource counts do not always provide equal service because users may have different channel gains and traffic requirements. A channel-aware scheduler may increase total throughput while repeatedly favouring users with strong signals. A fairness-aware scheduler can use historical throughput, minimum-rate constraints or priority classes to prevent starvation. The appropriate fairness definition depends on the service. Emergency voice, ordinary browsing and periodic telemetry may require different priority rules, so one global fairness score may not describe every operational need.

Channel Allocation and Efficiency

The score calculation for OFDMA demonstrates the weighted-sum method. Each rating is multiplied by the corresponding weight, and the five contributions are added to obtain a score out of five. The same calculation should be performed for every technique using the same weights. A table of scores should also show the underlying ratings so that the result can be audited. Reporting only the final number hides whether a technique scored well because of spectral efficiency, scalability or another criterion. The explanation is especially important when two methods receive similar overall scores.

Sensitivity analysis checks whether the ranking changes when the design priorities change. If a small adjustment to the weights reverses the result, the decision is sensitive and should be treated with caution. If a method remains near the top over a reasonable range of weights, the choice is more robust. The test should vary one weight at a time and also consider plausible combined changes. The weights should be renormalised to sum to one after each change so that scores remain comparable. This analysis reveals which assumptions have the greatest influence on the recommendation.

Capacity and Resource Utilisation

The four deployment cases illustrate why a hybrid approach is often necessary. Wide-area voice needs predictable access and continuity; a tactical link may prioritise robustness and resistance to interference; a stadium downlink needs high aggregate capacity under concentrated demand; and thousands of sensors need efficient access for short, infrequent reports. One technique may form the main resource structure while another improves performance. For example, OFDMA can provide frequency scheduling while SDMA reuses resources across spatial streams. Shared scheduled access can support sensors without reserving a permanent channel for every device.

The final selection should connect the weighted scores to the operating requirements of each deployment. A high score is meaningful only if the technique can meet the required delay, reliability, coverage and capacity. The report should identify any assumptions, explain why a hybrid is required and state the circumstances in which the recommendation would change. A sensitivity result should be used to identify risks rather than to claim certainty. This approach produces a defensible engineering decision and recognises that real communication systems combine multiple techniques to balance competing requirements.

Data Synchronisation and Consistency

The multiple-access comparison should distinguish legacy access methods from the way modern systems combine them. A modern radio interface may use OFDMA in the time-frequency domain and multiple antennas for spatial multiplexing. The system may also use coding, retransmission and scheduling to improve reliability. It is therefore more accurate to describe a deployment by its complete radio-resource structure than to label it with only one access technique. The assignment compares the principles so that the strengths and limitations of each dimension can be understood.

FDMA provides frequency separation and can make resource assignment easy to understand, but a fixed channel may remain unused when its assigned user is inactive. TDMA allows a carrier to be shared by users in different slots, but it requires timing and synchronisation. CDMA allows simultaneous code-separated transmissions, but interference depends on correlation and received power. OFDMA provides flexible subcarrier scheduling, but it needs accurate frequency and timing synchronisation. SDMA reuses the same resource across spatially separable users, but it requires antenna arrays and channel estimation. These are complementary rather than mutually exclusive capabilities.

Simulation Design and Validation

The scoring scale should be defined before ratings are assigned. For example, a rating of one may mean that the technique performs poorly for the stated criterion under the deployment assumptions, while a rating of five means that it performs very well. Intermediate ratings should have consistent meanings. Two people using the same definitions should be able to produce broadly similar scores. If the score is based on measured simulation results, the measurement should be cited in the explanation. If it is based on engineering judgement, it should be clearly labelled as a qualitative assessment.

The weights reflect the importance of the application rather than the inherent value of a technology. A stadium downlink may give high weight to spectral efficiency and scalability, while a tactical link may give more weight to robustness and continuity. A low-rate sensor

network may place greater importance on battery life and the ability to sleep between transmissions. The matrix can be adapted for each deployment, but the criteria and ratings must remain transparent. Using one set of weights for every application can conceal important differences in service requirements.

Simulation Design and Validation

A high spectral-efficiency rating should be checked against the actual bandwidth and coding assumptions. A technique may appear efficient because the model excludes control overhead or retransmissions. A fair comparison should calculate net payload throughput after subtracting guard intervals, pilots, control messages and protocol headers. The same accounting boundary should be used for all methods. If the simulation reports physical-layer rate for one method and application payload rate for another, the scores are not directly comparable.

Latency should be evaluated as a distribution rather than only an average. A system may have a low mean delay but occasional very long delays during congestion. For voice, interactive control and emergency messages, the 95th or 99th percentile may be more important than the mean. The model should identify whether delay includes waiting for a resource, transmission time, retransmission and processing. For telemetry, the reporting deadline can be used to calculate the percentage of records delivered on time. This creates a direct link between the decision matrix and the application requirement.

Data Synchronisation and Consistency

Scalability should include control-plane load and device complexity. As the number of users grows, the network may need more scheduling decisions, feedback reports, code assignments or handover procedures. The signalling load can become a bottleneck even when the radio payload capacity appears sufficient. A large population of low-power devices may also have limited processing and memory. The evaluation should therefore consider how the technique behaves as the user count increases and whether the devices can implement the required synchronisation and access procedures efficiently.

The tactical-link case requires careful wording about resistance to jamming. Spread-spectrum techniques can make some forms of interference more difficult to create or exploit, but they do not make a radio link immune to jamming. A sufficiently powerful or well-designed interferer can still disrupt reception. The system should use appropriate security, frequency agility, authentication and interference monitoring where permitted. The assignment should distinguish resistance to narrowband interference from protection against deliberate wideband jamming, because these are different threat models.

Link Availability and Offline Operation

The sensor deployment should consider the cost of synchronisation and wake-up signalling. A scheduled system can save energy if devices sleep until their assigned transmission time, but it must keep clocks sufficiently accurate or periodically resynchronise. A contention-based system avoids permanent scheduling for every device but may consume energy through repeated listening and collisions. Batching reports can reduce overhead, but it increases the time before each measurement is delivered. The best design depends on whether the application values immediate reporting, long battery life or efficient use of shared spectrum.

The weighted matrix should be treated as a decision aid that supports, rather than replaces, engineering analysis. The final recommendation should refer to the calculations and experiments in Q2 through Q5. It should explain why the selected technique suits each deployment, identify the main trade-off and state what additional test would be needed before installation. If two techniques have similar scores, cost, interoperability, equipment availability and maintenance capability can be used as tie-breakers. The final choice should remain traceable to the requirements and evidence.

Capacity and Resource Utilisation

The comparison should also include implementation maturity and interoperability. A theoretically efficient access method may be difficult to deploy if compatible devices, base-station equipment or operational expertise are unavailable. A widely supported method may have a lower theoretical score but a lower deployment risk and lower maintenance cost. The decision matrix can include an implementation-cost criterion or discuss it separately after the technical scoring. The final recommendation should explain whether the selected method is practical within the organisation's budget and infrastructure constraints.

The weight of fairness should be interpreted in relation to the service model. Equal throughput may be appropriate for a group of similar users, but emergency services may require priority based on urgency rather than equal sharing. A sensor network may prioritise alarms over routine periodic readings. A public broadband service may use proportional fairness to balance user experience and system throughput. The score should therefore reflect the intended service policy and not assume that equal allocation is always the fairest outcome.

Capacity and Resource Utilisation

A hybrid design can reduce the weaknesses of individual methods. OFDMA can divide resources into schedulable subcarriers, while SDMA can allow the same subcarriers to be reused by users with sufficiently distinct spatial channels. Coding and retransmission can improve reliability, while scheduling and admission control protect latency and fairness. The resulting system is more complex than any one method in isolation, so the benefits should be compared with added signalling, processing and equipment costs. A hybrid is justified when its measured gains are meaningful for the application.

The sensitivity analysis should include a record of the ranking under every tested weight set. If OFDMA remains the preferred method for the stadium case when spectral-efficiency weight is reduced and fairness weight is increased, the choice is relatively robust. If CDMA becomes preferred for the tactical case only when interference robustness receives a high weight, the result should be explained as a requirement-driven decision. A simple table of rank changes helps the reader see which recommendations depend strongly on subjective weighting.

Deployment Cost and Practicality

The final table should avoid mixing qualitative ratings with measured values without explanation. A rating of five for scalability is a judgement unless it is supported by a defined test such as maximum user count at a specified blocking probability. If measured evidence is available, it should be shown alongside the rating. If not, the rating should be labelled as an

engineering assessment. This distinction protects the credibility of the comparison and makes it clear what further testing is required before procurement.
A decision matrix should be reviewed when the application requirements change. If the stadium deployment changes from ordinary browsing to emergency video, latency and reliability may receive higher weights. If a sensor deployment becomes battery-powered and remote, energy consumption may become a major criterion. The original ranking should not be reused without checking whether the priorities remain valid. This makes the method adaptable and prevents a technically correct recommendation for one scenario from being applied incorrectly to another.

Simulation Design and Validation

The final report should identify which criteria are based on direct measurements and which are based on engineering judgement. For example, spectral efficiency can be calculated from measured net throughput and bandwidth, while implementation maturity may be assessed qualitatively. Keeping these categories separate makes the scoring more transparent. It also identifies where further experiments or vendor information are needed before a real deployment decision can be made.
A complete recommendation should include a fallback plan. If the preferred method fails to meet its target under congestion, the network may reserve more resources, reduce non-essential traffic, change the scheduling policy or use a complementary access method. A fallback should be designed before deployment rather than improvised during an outage. The assignment's comparison is therefore not only about selecting the highest-scoring method; it is about understanding how the network will continue to operate when assumptions are not met.

Simulation Design and Validation

The final comparison should be reproducible by another reader. The report should provide the criterion weights, rating definitions, individual ratings, weighted calculations and sensitivity cases. If the ratings change after a simulation is completed, the reason for the change should be documented. This makes the selection process transparent and allows the recommendation to be updated when new evidence becomes available.
A technology choice should also consider the migration path from existing infrastructure. Replacing every device and base station may be more expensive than introducing a new access method gradually. A hybrid solution can allow legacy voice or sensor equipment to continue operating while newer equipment uses more efficient scheduling. The deployment plan should include compatibility, training, maintenance and staged testing so that the technical improvement can be achieved without disrupting essential services.

ANSWER – Q7: Day-4 Design Challenge – Diagnose, Redesign and Validate

Selected limitation: Dedicated FDMA/TDMA channels waste capacity when telemetry traffic is bursty.

A. Diagnose the bottleneck (2 marks)

A dedicated channel is reserved even when a sensor has no report to send.

For a 10% duty-cycle source, the channel is idle for approximately 90% of the time.

As the number of sensors increases, the number of reserved channels grows even though only a small subset is transmitting at any moment.

The limitation is fixed resource reservation rather than the radio bandwidth alone.

B. Define the design objective (2 marks)

Objective: reduce idle reserved resource units by at least 50% during bursty traffic while delivering at least 99% of generated reports within the selected reporting deadline. The deadline and delivery target are design assumptions and should be checked against the application requirements.

C. Revised strategy (4 marks)

Use a shared scheduled pool. Sensors keep reports locally, request or receive a short transmission opportunity, send the report, and release the resource immediately. Batch multiple readings when permitted. Add sequence numbers and acknowledgements so retries do not create duplicate records.

Pseudocode

```
for each reporting interval:
    collect reports from sensors
    place each report in a local queue with a sequence number
    announce or assign available shared transmission slots
    while slots remain and queued reports exist:
        select a queued report
        transmit report in an assigned slot
        if acknowledgement received:
            remove report from queue
        else:
            retain report for retry
    record delivered, delayed, lost, occupied_slots, and idle_slots
```

Operations reduced: permanent reservation for silent devices and idle channel holding. Additional costs: scheduling messages, queue memory, acknowledgement overhead, and possible delay while waiting for a slot. Dedicated channels may still be preferable for continuous voice or strict low-latency services.

D. Prediction before implementation (1 mark)

Prediction: under 10% duty-cycle traffic, shared scheduling should reduce reserved-but-idle resource units by more than 50%, while maintaining delivery through queueing. It may add a small scheduling delay compared with an immediately available dedicated channel.

E. Illustrative validation calculation (2 marks)

Queue Management and Recovery

Assume 100 resource units are available per reporting interval and 10 sensors are active on average. In a dedicated design, if 100 sensors each reserve one unit, 100 units are reserved and 90 are idle. In a shared design, 10 units are occupied by the active reports; if 2 extra units are used for scheduling/control, total used is 12.

Measure	Original dedicated allocation	Redesigned shared pool
Reserved/used units	100 reserved	10 data + 2 control = 12 used
Idle data units	90	0 of the 10 data slots assigned; remaining capacity stays available to other traffic
Data utilisation against 100 units	10/100 = 10%	10/100 = 10% payload; only 12% including control
Resource reservation reduction	Baseline	(100 − 12) ÷ 100 × 100 = 88% fewer reserved/used units
Report delivery	Depends on assigned channel availability	Queued reports delivered when slots and acknowledgements are available

The 88% figure is an illustrative arithmetic comparison under the stated assumptions, not a measured test result. A proper validation run should use identical generated arrivals and fixed random seeds for both designs and export CSV columns for arrival time, delivery time, retries, slot count, blocked reports, and lost reports.

F. Defend the final design (2 marks)

The redesign addresses the root cause by sharing resources among intermittent senders rather than reserving a channel for each device. The predicted reduction in reservation is supported by the example calculation. The trade-off is extra scheduling and queueing delay; therefore, shared access is suitable for periodic telemetry, while dedicated or guaranteed slots remain useful for continuous voice and emergency traffic.

Simulation Design and Validation

The design challenge begins by identifying the actual bottleneck in the original arrangement. When a device reserves a dedicated channel but sends a report only occasionally, the resource remains unavailable to other devices during the idle period. The problem is not necessarily a shortage of total spectrum; it may be poor matching between the reservation method and the traffic pattern. The diagnosis should use measured occupancy over time and compare the number of reserved units with the number carrying useful payload. This distinction is essential before changing the access method.

A 10% duty-cycle device is active for only a small fraction of the reporting interval. If one resource unit is reserved continuously, approximately 90% of its reservation time carries no

payload. Across a large fleet, the unused reservations accumulate and can prevent other devices from accessing the network. The actual waste depends on report duration, protocol overhead, acknowledgement timing and whether the resource can be released immediately after transmission. The simulation should measure occupied payload slots, control slots, idle reserved slots and the total reservation duration separately.

Simulation Design and Validation

The revised objective should be measurable. Reducing idle resource reservation by at least 50% is one target, while delivering at least 99% of generated reports within a stated deadline is a separate service target. Both must be tested because a design could reduce idle reservations simply by delaying or dropping reports. The deadline should be selected from the utility application requirements, not chosen only to make the simulation appear successful. If some readings are safety-critical, they may require a shorter deadline or a higher priority than routine meter data.

A shared scheduled pool allows devices to request or receive short transmission opportunities when they have data. A sensor can store a report locally, wait for an assigned slot, transmit the report and release the resource. The scheduler can combine requests, assign priorities and limit the number of simultaneous transmissions. Sequence numbers help identify repeated reports, while acknowledgements confirm successful receipt. If an acknowledgement is not received, the device retains the report and retries according to a backoff rule. This avoids treating every temporary failure as permanent data loss.

Queue Management and Recovery

The scheduler must prevent one device from occupying the shared pool indefinitely. A fair policy can limit consecutive transmissions, use round-robin selection among waiting devices or prioritise reports according to age and importance. If a device repeatedly fails, the scheduler should avoid allowing its retries to block all other devices. Backoff timers can reduce repeated simultaneous attempts after a collision or outage. The policy should be simple enough for low-power devices and should not require more signalling than the payload traffic it is intended to support.

The queue at each sensor must be sized for the longest expected outage and reporting rate. If reports are generated faster than they can be delivered, the queue grows until storage is exhausted. The design should define what happens when the queue reaches its limit: it may preserve the newest reading, preserve the oldest unsent reading, aggregate readings or raise an error. For meter data, losing a historical record may be unacceptable, so local storage and recovery procedures should be included in the design. Queue length and overflow events must be part of the test output.

Simulation Design and Validation

A robust simulation should generate identical arrival times for the dedicated and shared designs. The same random seed, report sizes and outage pattern should be used so that differences are caused by the access strategy rather than by different traffic samples. Record arrival time, request time, assigned slot, transmission time, acknowledgement time, retries, delivery time and final status for each report. From these records, calculate delivery ratio,

delay distribution, resource utilisation, idle reservation and retransmission overhead. This event-level data allows the result tables and graphs to be independently checked.
The illustrative calculation with 100 available resource units assumes that 100 devices each reserve one unit in the dedicated design, while only 10 are active during the interval. The shared design uses 10 data units and two control units, for a total of 12 used units. The difference of 88 units represents an 88% reduction in reserved or used units under those simplified assumptions. It is not a measured network result and does not include all possible scheduling overheads, retransmissions or collision costs. A real experiment must replace the assumed counts with measured simulator outputs.

Simulation Design and Validation

The shared design may increase access delay because a device has to wait for a grant or an available contention opportunity. This delay can be acceptable for periodic meter reports but may be unsuitable for emergency voice or strict real-time control. The system can support different service classes by reserving guaranteed resources for urgent traffic and allowing routine telemetry to use the shared pool. Admission control should protect the high-priority service from overload. The redesign is therefore a targeted improvement for bursty telemetry, not a universal replacement for every dedicated radio resource.
The final defence should compare the original and redesigned systems using the same traffic and channel assumptions. The recommendation is justified if it reduces idle reservation, maintains the delivery target, keeps delay within the application deadline and does not create excessive control overhead. The report should also state the limits of the experiment, including simplified propagation, assumed report sizes and any unmodelled device failures. If the shared system fails the deadline or reliability target during a large burst, the next step may be to add scheduled capacity or stagger reporting times rather than returning to permanent reservation for every device.

Simulation Design and Validation

The redesign should define the reporting deadline before testing. A report delivered after the deadline may be useful for historical analysis but may not satisfy an operational requirement. For example, a routine meter reading may tolerate several minutes of delay, while an alarm indicating equipment overheating may need immediate delivery. The scheduler should support priority classes or separate queues so that urgent reports are not delayed behind routine data. The experiment should report on-time delivery separately for each class rather than combining all records into one average.
A shared pool needs a mechanism for devices to obtain access. One option is a request-and-grant process in which a sensor sends a short request and the base station assigns a data slot. Another option is contention-based access in which devices select a random opportunity and retry after a collision. Request-and-grant reduces collisions but adds control signalling and an initial request delay. Contention-based access can be simple for small loads but may perform poorly during synchronised bursts. The design should select the access procedure based on device capability and expected traffic density.

Simulation Design and Validation

The backoff rule should prevent repeated collisions when many devices attempt to transmit at once. A binary exponential backoff can increase the random waiting window after each failed attempt, reducing the chance that devices retry together. However, a large backoff window increases delay and may cause reports to miss their deadline. The simulation should compare at least two backoff settings and report collision count, retry count, average delay and 95th-percentile delay. The chosen rule should balance rapid recovery with stable access under heavy load.

A scheduler can use report age to prevent starvation. If it always selects the newest or highest-rate device, older reports may remain in the queue for too long. An oldest-first policy serves the longest-waiting report, while a priority-weighted policy can give urgent alarms precedence. A maximum-wait limit can trigger a report even when its normal priority is low. These policies should be tested with mixed traffic, because a scheduler that performs well for identical periodic reports may behave differently when alarms and routine readings share the same channel.

Data Synchronisation and Consistency

The control overhead must be included in the resource-efficiency calculation. A shared design may save many idle data slots but use additional slots for requests, grants, acknowledgements and synchronisation. The net saving is the difference between the original reserved resources and the total data plus control resources used by the redesign. If control overhead grows too large during a burst, the expected gain may disappear. The simulator should count each control message and convert its transmission time into resource units so that the comparison uses a consistent measure.

A practical implementation should define recovery behaviour after a base-station or gateway restart. Devices may all attempt to reconnect immediately, creating a second burst. A randomised reconnect delay can spread the requests over time. The gateway should also retain enough state to identify previously delivered reports and reject duplicates. If the acknowledgement was lost after the server accepted a report, the device may retransmit it; a unique record identifier allows the server to recognise the duplicate without losing the original measurement.

Queue Management and Recovery

The validation should include a normal operating period, a short burst, a long outage followed by reconnection and a high-priority alarm during congestion. Each scenario tests a different aspect of the redesign. Normal operation checks routine efficiency; a burst checks contention and scheduling; an outage checks queue capacity and recovery; and an alarm checks priority handling. The same event schedule should be applied to the original and redesigned systems. Results should be presented in a table with clear units and a separate graph for delay and resource use.

A useful graph plots cumulative reports generated and cumulative reports delivered over time. The distance between the curves shows the backlog, while the slope of the delivery curve shows the rate at which the system clears queued reports. If the delivered curve catches up after a burst, the system has recovered. If the gap continues to grow, the offered load exceeds the sustained service rate. This graph is often more informative than a single average

throughput value because it reveals whether the system can drain its queue after a temporary overload.

Queue Management and Recovery

The design should include a stability check. For a queue to remain bounded over the long term, the average service rate must exceed the average arrival rate, allowing for protocol overhead and retransmissions. If reports arrive faster than the system can transmit them, no queue size can prevent eventual overflow. Temporary bursts can be handled with buffering, but sustained overload requires more capacity, admission control or a lower reporting rate. The report should distinguish a short overload that can be buffered from a permanent mismatch between offered load and service capacity.

The final recommendation should identify the conditions under which the shared scheduled pool remains appropriate. It is suitable when reports are short, traffic is intermittent, delay tolerance exists and the system can coordinate access reliably. Dedicated resources remain justified for continuous voice, emergency services or applications with strict deterministic delay. A hybrid arrangement can protect urgent traffic while sharing resources among routine sensors. The conclusion should be based on measured delivery, delay, utilisation and overhead, and should clearly separate illustrative calculations from results produced by the actual simulator.

Simulation Design and Validation

The deployment should be monitored after installation because real traffic may differ from the generated simulation. The operator can collect counts of access requests, successful grants, collisions, retransmissions, queue depth and delivery delay. These measurements should be aggregated without exposing unnecessary personal or device information. If the measured system repeatedly approaches its queue or delay limits, the scheduler can adjust reporting windows or resource allocation. Monitoring turns the redesign into an operational process rather than a one-time calculation.

The reporting schedule can be adapted based on the state of the network. During normal operation, devices may report at their standard interval. During congestion, non-urgent reports can be delayed within an allowed window, while alarms remain prioritised. After an outage, the gateway can request devices to reconnect in groups rather than allowing every device to transmit immediately. Such controlled recovery reduces the chance of a second overload and helps the network return to normal operation more smoothly.

Data Synchronisation and Consistency

The system should protect against repeated delivery of the same record. A sensor may transmit a report successfully but fail to receive the acknowledgement. It will then retry, and the server may receive the same payload more than once. The unique record identifier allows the server to recognise the duplicate and acknowledge it without creating a second database entry. The test should count duplicate transmissions separately from duplicate accepted records, because retransmission is a normal reliability mechanism while duplicate storage is an application error.

The cost comparison should include both capital and operating expenses. A shared access design may reduce the need for dedicated radio resources, but it may require a scheduler,

additional software, device firmware changes and monitoring tools. It may also require more testing during commissioning. The original dedicated design may be simpler but waste capacity as the device population grows. A fair decision should estimate the cost over the expected operating life and include maintenance, energy and future expansion rather than comparing only initial equipment prices.

Queue Management and Recovery

A sensitivity test should vary the number of active devices, report size, reporting interval, outage duration and control overhead. If the redesign performs well only when 10% of devices are active, the report should identify the point at which the shared pool becomes overloaded. The break-even point can be expressed as the offered load or active-device count at which the shared system no longer meets the delay or delivery target. This gives the operator a practical threshold for adding capacity or changing the reporting schedule.

The operator should define a trigger for expanding the shared pool. For example, expansion may be considered when the 95th-percentile queue delay exceeds the reporting deadline for several consecutive measurement periods or when queue occupancy repeatedly crosses a defined threshold. A trigger based on repeated measurements is more reliable than reacting to a single temporary burst. The trigger should also consider whether the overload is local to one gateway or affects the entire network, because the appropriate remedy may differ.

Simulation Design and Validation

The redesign should be tested for fairness between devices. A device with a stronger radio link may complete transmissions quickly and use fewer retries, while a device at the edge of coverage may need more resources. If the scheduler always favours the quickest transmission, weak devices may remain in the queue. A fair policy can limit repeated service to one device or increase priority with waiting time. The experiment should report the distribution of delivery delay across devices, not only the network-wide average.

The final implementation should preserve a record of configuration changes. If reporting intervals, backoff windows or scheduling priorities are modified, the operator should record the old and new values, the reason for the change and the measured effect. This makes it possible to compare performance over time and reverse a change that causes unexpected behaviour. Configuration management is part of validation because a result cannot be reproduced if the operating parameters are unknown.

Simulation Design and Validation

The design should be considered successful only when it meets the stated targets over repeated test runs and representative traffic conditions. A single favourable run is not enough to demonstrate reliability. The final evidence should include raw event logs, summary tables, graphs, random seeds and the exact simulation parameters. It should also identify any scenario where the target was not met. Reporting limitations honestly is part of engineering validation and helps determine what must be improved before deployment.

The raw simulation data should be retained with a description of the software version, parameter values and random seed. Summary tables alone are not enough to reproduce a result because they do not show individual events or the order in which requests were processed. Keeping the event log allows the operator to investigate unusual delay, duplicate records or

queue overflow. It also provides evidence that the reported improvement came from the revised access strategy.

Simulation Design and Validation

The final defence should connect the redesign to the original diagnosis. If the original problem was idle reservation by intermittent devices, the evidence should show that the revised system reduces idle reservation while preserving delivery reliability and acceptable delay. If the results do not meet those conditions, the redesign should be adjusted rather than declared successful. A defensible conclusion follows from the measurements and acknowledges the remaining limits of the model.

Overall conclusion

The assignment shows that multiple-access design is a trade-off among capacity, reliability, fairness, delay, and complexity. Capacity calculations and simulations should state their assumptions, use fixed seeds, and report counted measures. Illustrative tables in this document demonstrate the calculation method; they should not be represented as actual measured results unless the corresponding simulations are run and the outputs are recorded.

Mark Distribution

Question	Assessed Area	Marks
Q1	Mobile communication vs. mobile computing	5
Q2	FDMA and TDMA	6
Q3	CDMA	7
Q4	OFDMA	6
Q5	SDMA	6
Q6	Comparison of SDMA, TDMA, FDMA, CDMA, OFDMA	7
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	13
TOTAL		50

Submission Requirements (Digital)

- Assignment report (PDF) containing the required evidence for Q1–Q7. No handwritten work.
- Source code for all implementations/simulations (GitHub repository link with README and fixed random seeds).
- Raw experimental data (CSV) used to produce your results.
- Tables and graphs used in your analysis.
- Counted quantities (slots, messages, bits, blocked calls, codes, throughput) are primary experimental evidence. Execution time may be reported as supporting evidence but must not be the sole basis for conclusions.
- Standards parameters (frame durations, rates, header sizes, etc.) must be obtained from cited references; state assumptions wherever a value is assumed.

Assessment Verification

You may be asked to analyse any part of your design, code, test-data design, experimental results, predictions, analysis and Design Challenge in a brief verification interpretion/viva. Disclose any AI or reference tools used. You must be able to analyse and defend the evidence as your own work.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook descriptions of mobile systems. It assesses your ability to:

Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate

Your conclusions and design decisions must be supported by your own reasoning and experimental evidence.